

Project Data Page:

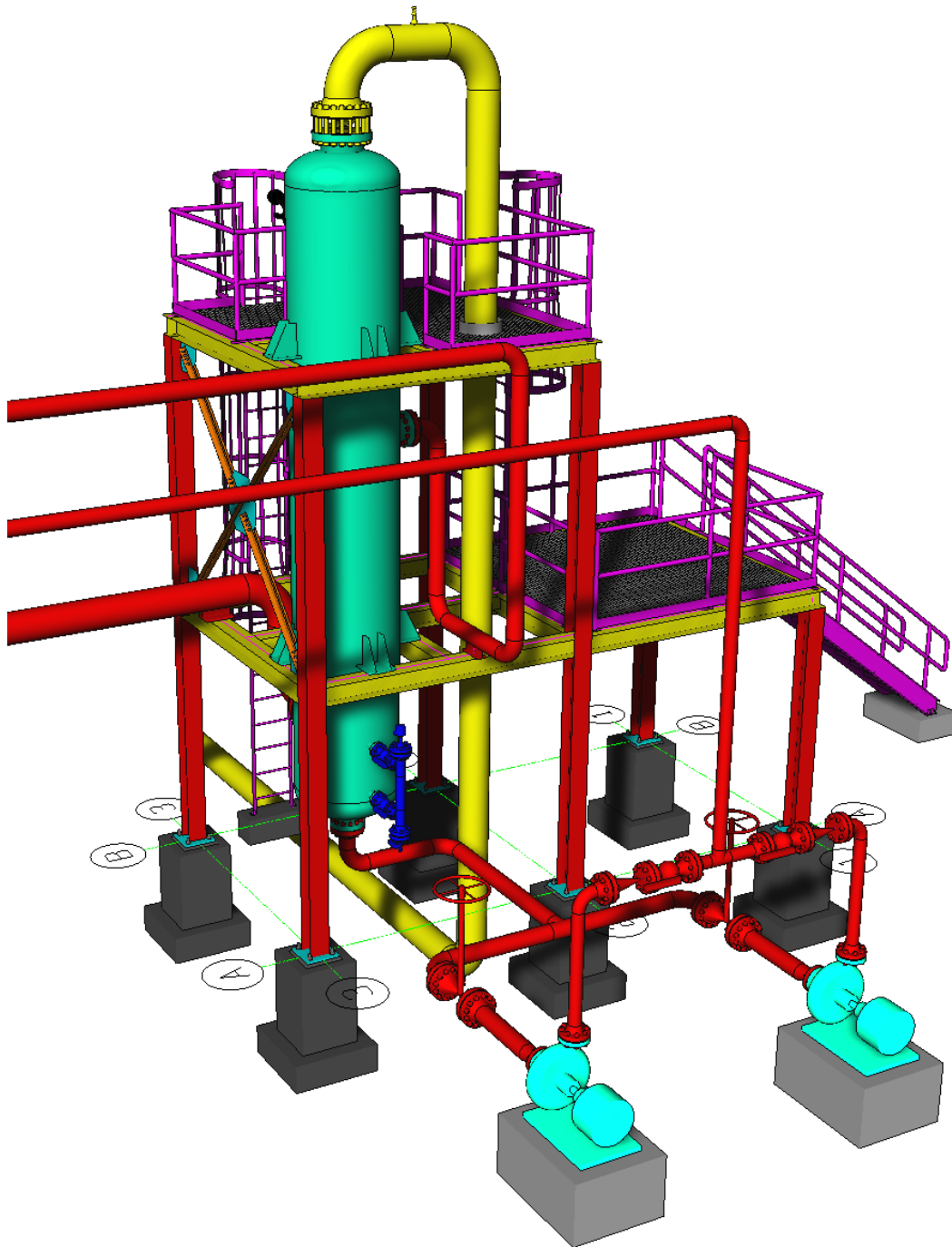




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\402-V-004 - Copy 7%.pvdb

Date of Analysis : May 26, 2026 5:47pm

PV Elite 28 U1, March 2026



Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1000.000 mm. ID
 Vessel Design Length, Tangent to Tangent 2700.00 mm.
 Specified Datum Line Distance 50.00 mm.
 Internal Design Temperature 65 °C
 Internal Design Pressure 28.500 bars
 External Design Temperature 0 °C
 Maximum Allowable Working Pressure 30 bars
 Shop Test Pressure 38.643 bars
 Field Test Pressure 38.643 bars
 Required Minimum Design Metal Temperature -38.0 °C
 Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Wind Design Code IS 875 (Part 3):2015 Amd. 1&2 2020
 Earthquake Design Code IS 1893 RSM (Part 1): 2024

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Nozzle	SA-516 70	...	...	K02700	
Nozzle	SA-350 LF2	1	...	K03011	+IT
Nozzle	SA-333 6	...	...	K03006	+IT
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-350 LF2	1	...	K03011	+IT
Nozzle Flg	SA-105	...	...	K03504	
Hrz Bolting	SA-285 C	...	...	K02801	
Saddles	SA-516 70	...	...	K02700	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Left D end	28.560	0.00	30.000	3.0000	No	No
Shell	28.560	0.00	30.000	3.0000	N/A	No
Right D end	28.560	0.00	30.000	3.0000	No	No

Liquid Level: 900.00 mm. Dens.: 0.001 kg./cm³ Sp. Gr.: 0.682

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Ellipse	0.0	50.0	18.0	14.0	13.4	...	1.00	1.00
Cylinder	2600.0	2600.0	14.0	14.0	13.5	...	1.00	1.00
Ellipse	2650.0	50.0	18.0	14.0	13.4	...	1.00	1.00



Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1_6"	WRC-297	0.949	Shell	Passed
N2_6"	WRC-297	0.949	Shell	Passed
N4_2"	WRC-297	0.949	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	125.000	mm.
Saddle Bearing Angle	128.000	deg.
Centerline Dimension	925.000	mm.
Wear Pad Width	180.000	mm.
Wear Pad Thickness	6.000	mm.
Wear Pad Bearing Angle	140.000	deg.
Distance from Saddle to Tangent	540.000	mm.
Baseplate Length	965.000	mm.
Baseplate Thickness	16.000	mm.
Baseplate Width	150.000	mm.
Number of Ribs (including outside ribs)	3	
Rib Thickness	16.000	mm.
Web Thickness	16.000	mm.
Height of Center Web	306.000	mm.
Number of Bolts in Baseplate	2	

Baseplate Sketch

Maximum Tensile Bolt Load	106.	Kgf
---------------------------	------	-----

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	5829.70	Kgf
Maximum Transverse Saddle Shear Load	892.12	Kgf
Maximum Longitudinal Saddle Shear Load	502.07	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	4357.33	Kgf
Maximum Transverse Saddle Shear Load	294.40	Kgf
Maximum Longitudinal Saddle Shear Load	308.08	Kgf

Weights:

Fabricated - Bare w/o removable internals	2566.8	kg.
Shop Test - Fabricated + water (full)	4948.2	kg.
Shipping - Fab. + rem. intls.+ shipping app.	3792.2	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	3792.2	kg.
Empty - Fab. + intls. + details + wghts.	3792.2	kg.
Operating - Empty + operating liquid (no ca)	5338.5	kg.
Field Test - Empty weight + test medium (full)	6173.6	kg.



Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
M1_20"	30	...	...	...		OK	Passed
N7A_2"	30	...	...	OK	7.80	OK	Passed
N7A_2"	30	...	...	OK	7.80	OK	No Calc[*]
N7B_2"	30	...	...	OK	7.80	OK	No Calc[*]
N3_1"	30	...	...	OK	6.12	OK	No Calc[*]
N1_6"	30	...	...	OK	9.22	OK	Passed
N2_6"	30	...	...	OK	9.22	OK	Passed
N4_2"	30	...	...	OK	7.80	OK	No Calc[*]
N5_1"	30	...	...	OK	6.12	OK	No Calc[*]
N8_6"	30	...	...	OK	9.22	OK	Passed
N9_2"	30	...	...	OK	7.80	OK	No Calc[*]
N10_2"	30	...	...	OK	7.80	OK	No Calc[*]
N11_4"	30	...	...	OK	8.26	OK	Passed
N12_4"	30	...	...	OK	8.26	OK	Passed
N6_1 1/2"	30	...	...	OK	6.42	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 30 Nozzle : N124"
 Minimum MAWP Shells/Flanges : 30 Element : Shell
 Minimum MAPnc Shells/Flanges : 38 Element : Shell

Computed Vessel M.A.W.P. : 30 bars

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
10	M1_20"	0.000	0.000	1028.000
20	N7A_2"	250.000	56.780	113.600
20	N7B_2"	250.000	270.000	113.600
20	N3_1"	210.000	90.000	63.400
20	N1_6"	490.000	90.000	310.145
20	N2_6"	2280.000	90.000	310.145
20	N4_2"	2580.000	90.000	113.600
20	N5_1"	2000.000	90.000	63.400
20	N8_6"	1410.000	90.000	310.145
20	N9_2"	2550.000	270.000	113.600
20	N10_2"	1110.000	90.000	113.600
20	N11_4"	1770.000	90.000	201.662
20	N12_4"	860.000	90.000	201.662
30	N6_1 1/2"	0.000	0.000	88.200

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N3_1"	1.000 in	LWN	LWN	1.630	8.000	...	...	498.48	600
N5_1"	2.000 in	LWN	LWN	1.630	8.000	...	...	500.26	300
N6_1 1/2"	1.500 in	Actual	LWN	2.287	10.000	...	...	236.28	300
N7A_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	323.47	300
N7B_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	496.21	300
N4_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	212.63	600
N9_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	222.16	300
N10_2"	2.000 in	Actual	LWN	3.320	16.764	...	...	212.63	300
N11_4"	4.000 in	120	WNF	4.500	11.125	180.00	14.00	433.95	300
N12_4"	4.000 in	120	WNF	4.500	11.125	180.00	14.00	433.95	300
N1_6"	6.000 in	80	WNF	6.625	10.973	280.00	14.00	425.10	300
N2_6"	6.000 in	80	WNF	6.625	10.973	280.00	14.00	425.10	300
N8_6"	6.000 in	80	WNF	6.625	10.973	280.00	14.00	425.10	300
M1_20"	20.000 in	Actual	WNF	20.945	12.000	900.00	14.00	194.16	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N3_1"	SA-350 LF2	14.000	10.000	...	...	...
N5_1"	SA-350 LF2	14.000	10.000	...	...	...
N6_1 1/2"	SA-350 LF2	14.000	10.000	...	...	...
N7A_2"	SA-350 LF2	14.000	10.000	...	...	...
N7B_2"	SA-350 LF2	14.000	10.000	...	...	...
N4_2"	SA-350 LF2	14.000	10.000	...	...	...
N9_2"	SA-350 LF2	14.000	10.000	...	...	...
N10_2"	SA-350 LF2	14.000	10.000	...	...	...
N11_4"	SA-333 6	14.000	10.000	12.000	14.000	...
N12_4"	SA-333 6	14.000	10.000	12.000	14.000	...
N1_6"	SA-333 6	14.000	10.000	12.000	14.000	...
N2_6"	SA-333 6	14.000	10.000	12.000	14.000	...
N8_6"	SA-333 6	14.000	10.000	12.000	14.000	...
M1_20"	SA-516 70	14.000	10.000	12.000	14.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N3_1"	160.000	90.0	483.47	0.00	Shell
N5_1"	1950.000	90.0	485.25	0.00	Shell
N6_1 1/2"	...	0.0	207.96	0.00	Right D end
N7A_2"	200.000	56.8	206.38	0.00	Shell
N7B_2"	200.000	270.0	480.43	0.00	Shell
N4_2"	2530.000	90.0	196.85	0.00	Shell
N9_2"	2500.000	270.0	206.38	0.00	Shell
N10_2"	1060.000	90.0	196.85	0.00	Shell
N11_4"	1720.000	90.0	416.67	0.00	Shell



N12_4"	810.000	90.0	416.67	0.00	Shell
N1_6"	440.000	90.0	403.97	0.00	Shell
N2_6"	2230.000	90.0	403.97	0.00	Shell
N8_6"	1360.000	90.0	403.97	0.00	Shell
M1_20"	...	0.0	140.47	0.00	Left D end



Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm. THK X 1000.0mm. ID X 50.0mm.	SA-516 70
1	INSULATION: 300mm X 25mm THK	PP
2	SADDLE: 125.0mm X 128 DEG	SA-516 70
1	PLATFORM: 1200mm X 2000mm WIDE	
1	LADDER: 1655.0mm	...
1	INSULATION: 2600mm X 25mm THK	PP
1	INSULATION: 250mm X 25mm THK	PP
1	CLASS 300 GR 1.1, 20.0" BLIND FLANGE(S)	SA-350 LF2
2	CYLINDRICAL SEGMENT 14.0mm. THK X 1570.8mm. X 2600.0mm.	SA-516 70
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Left D end	[10]	B	-35	-36		0.981	14.000	1.00	No
Left D end	[7]	B	-27	-42	-29	0.733	18.000	1.00	No
Shell	[8]	B	-18	-18	-29	1.000	14.000	1.00	No
Right D end	[10]	B	-35	-36		0.981	14.000	1.00	No
Right D end	[7]	B	-27	-42	-29	0.733	18.000	1.00	No
Fixed saddle	[24]	B	-29	-29	-29	1.000	6.000	1.00	No
New Saddle	[24]	B	-29	-29	-29	1.000	6.000	1.00	No
M120"	[1]	B	-35	-38		0.940	14.000	1.00	No
Nozzle Flg	[5]	B	-46	-48					
N7A2"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N7B2"	[1]	B	-35	-37		0.958	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N31"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N16"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-72					
N26"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-72					
N42"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N51"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N86"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-72					
N92"	[1]	B	-35	-37		0.958	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N102"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-46					
N114"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-72					
N124"	[1]	B	-35	-37		0.956	14.000	1.00	No
Nozzle Flg	[5]	!	-46	-72					
N61 1/2"	[1]	B	-46	-104		0.066	10.000	1.00	No
Nozzle Flg	[5]	A	-18	-44					
Nozl. Bolting	[22]		-48			1.000	64.000	1.00	No

Warmest MDMT: -18 -18

Required Minimum Design Metal Temperature -38.0 °C

Warmest Computed Minimum Design Metal Temperature -29.0 °C

Warning: Computed overall MDMT was higher than the required value !

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations



[21] - Body Flange Bolting Material
[22] - Nozzle Flange Bolting Material
[23] - Stiffening Ring to Shell Weld
[24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (barsUniCode):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm2
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm2
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm3
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm3
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm2
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m2
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm3
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	28.5	bars
Design Internal Temperature	65.0	°C
Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Minimum Design Metal Temperature	-38.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000



Consider Vortex Shedding

N

Shop Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition No

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	
Damping Factor (Beta) for Wind (Empty)	0.0000	
Damping Factor (Beta) for Wind (Filled)	0.0000	

Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117



0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/3
Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	Left D end
Distance "FROM" to "TO"	50 mm.
Inside Diameter	1000 mm.
Element Thickness	14 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	18 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	28.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm2
Allowable Stress, Operating	138 N./mm2
Allowable Stress, Hydrotest	234 N./mm2
Material Density	0.00775 kg./cm3
P Number Thickness	32.004 mm.
Yield Stress, Operating	246 N./mm2
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Elliptical Head Factor	2.0



Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	900 mm.
Liquid Density	0.0006814 kg./cm3
Element From Node	10
Detail Type	Insulation
Detail ID	Insul.0
Dist. from "FROM" Node / Offset dist	-250 mm.
Height/Length of Insulation	300 mm.
Thickness of Insulation	25 mm.
Density	0.00022 kg./cm3
Element From Node	10
Detail Type	Nozzle
Detail ID	M120"
Dist. from "FROM" Node / Offset dist	0 mm.
Nozzle Diameter	20 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	Y
Weight of Nozzle (Used if > 0)	461.53 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-516 70

Element#	2/3
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	2600 mm.
Inside Diameter	1000 mm.
Element Thickness	14 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	28.5 bars
Design Temperature Internal Pressure	65 °C
Design External Pressure	0 bars
Design Temperature External Pressure	0 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	20
Detail Type	Saddle
Detail ID	Fixed saddle
Dist. from "FROM" Node / Offset dist	490 mm.
Width of Saddle	125 mm.
Height of Saddle at Bottom	925 mm.
Saddle Contact Angle	128.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	180 mm.
Thickness of Wear Plate	6 mm.
Contact Angle, Wear Plate (degrees)	140.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.



Element From Node	20	
Detail Type	Saddle	
Detail ID	New Saddle	
Dist. from "FROM" Node / Offset dist	2210	mm.
Width of Saddle	125	mm.
Height of Saddle at Bottom	925	mm.
Saddle Contact Angle	128.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	180	mm.
Thickness of Wear Plate	6	mm.
Contact Angle, Wear Plate (degrees)	140.0	
Friction coefficient	0.1	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	20	
Detail Type	Platform	
Detail ID	PLAT 1	
Dist. from "FROM" Node / Offset dist	0	mm.
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	51271	cm2
Platform Weight	1081.8	Kgf
Platform Railing Weight	0	Kgf/mm.
Platform Grating Weight	0.017	Kgs/cm2
Platform Width	2000	mm.
Platform Height	1200	mm.
Platform Clearance or End Offset	230	mm.
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm.
Ladder End Elevation	1655	mm.
Unit Weight of Ladder	0.037	Kgf/mm.
Platform Length (top head platform)	3000	mm.

Element From Node	20	
Detail Type	Liquid	
Detail ID	Liquid 20	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	900	mm.
Liquid Density	0.0006814	kg./cm3

Element From Node	20	
Detail Type	Insulation	
Detail ID	Insul.0	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	2600	mm.
Thickness of Insulation	25	mm.
Density	0.00022	kg./cm3

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N7A2"	
Dist. from "FROM" Node / Offset dist	200	mm.
Nozzle Diameter	2	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	56.7803	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	10.356	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N7B2"	



Dist. from "FROM" Node / Offset dist 200 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 270.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 10.356 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N31"
Dist. from "FROM" Node / Offset dist 160 mm.
Nozzle Diameter 1 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 4.5303 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N16"
Dist. from "FROM" Node / Offset dist 440 mm.
Nozzle Diameter 6 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 41.259 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N26"
Dist. from "FROM" Node / Offset dist 2230 mm.
Nozzle Diameter 6 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 41.259 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N42"
Dist. from "FROM" Node / Offset dist 2530 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 600
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 10.819 Kg
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N51"
Dist. from "FROM" Node / Offset dist 1950 mm.
Nozzle Diameter 1 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 90.0



Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 4.3746 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N86"
Dist. from "FROM" Node / Offset dist 1360 mm.
Nozzle Diameter 6 in.
Nozzle Schedule 80
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 41.259 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N92"
Dist. from "FROM" Node / Offset dist 2500 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 270.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 10.356 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N102"
Dist. from "FROM" Node / Offset dist 1060 mm.
Nozzle Diameter 2 in.
Nozzle Schedule None
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 6.3455 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-350 LF2 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N114"
Dist. from "FROM" Node / Offset dist 1720 mm.
Nozzle Diameter 4 in.
Nozzle Schedule 120
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 0 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT

Element From Node 20
Detail Type Nozzle
Detail ID N124"
Dist. from "FROM" Node / Offset dist 810 mm.
Nozzle Diameter 4 in.
Nozzle Schedule 120
Nozzle Class 300
Layout Angle 90.0
Blind Flange (Y/N) N
Weight of Nozzle (Used if > 0) 24.922 Kgf
Grade of Attached Flange GR 1.1
Nozzle Matl SA-333 6 +IT



Element From Node 20
 Detail Type Weight
 Detail ID clip Wt
 Dist. from "FROM" Node / Offset dist 1450 mm.
 Miscellaneous Weight 55 Kg
 Offset from Element Centerline 0 mm.

Element From Node 20
 Detail Type Weight
 Detail ID Stilling well N
 Dist. from "FROM" Node / Offset dist 810 mm.
 Miscellaneous Weight 30 Kg
 Offset from Element Centerline 0 mm.

Element From Node 20
 Detail Type Weight
 Detail ID Dip pipe (N4)
 Dist. from "FROM" Node / Offset dist 2530 mm.
 Miscellaneous Weight 10 Kg
 Offset from Element Centerline 0 mm.

Element From Node 20
 Detail Type Weight
 Detail ID Lifting attach.0
 Dist. from "FROM" Node / Offset dist 1300 mm.
 Miscellaneous Weight 80 Kg
 Offset from Element Centerline 0 mm.

Element# 3/3
 Element From Node 30
 Element To Node 40
 Element Type Elliptical
 Description Right D end
 Distance "FROM" to "TO" 50 mm.
 Inside Diameter 1000 mm.
 Element Thickness 14 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 18 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 28.5 bars
 Design Temperature Internal Pressure 65 °C
 Design External Pressure 0 bars
 Design Temperature External Pressure 0 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 70
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 30
 Detail Type Liquid
 Detail ID Liquid 30
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 900 mm.
 Liquid Density 0.0006814 kg./cm3

Element From Node 30
 Detail Type Insulation
 Detail ID Insul
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 250 mm.
 Thickness of Insulation 25 mm.
 Density 0.00022 kg./cm3

Element From Node 30
 Detail Type Nozzle
 Detail ID N61 1/2"



Dist. from "FROM" Node / Offset dist	360	mm.
Nozzle Diameter	1.5	in.
Nozzle Schedule	None	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6.1533	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-350 LF2 +IT	

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Left D end		50	...	50	...
Shell		2650	...	2600	...
Right D end		2700	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Left D end		28.56	18	3	1000	138	1
Shell		28.56	14	3	1000	138	2
Right D end		28.56	18	3	1000	138	3

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Left D end		28.5	30.2891	38.5292	14	13.3502	1
Shell		28.5	29.7254	37.9987	14	13.5418	2
Right D end		28.5	30.2891	38.5292	14	13.3502	3

Minimum 30 38

MAWP: 30 bars, limited by: Shell.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Left D end

Elliptical Head Design Information:					
Design Pressure	P	28.560 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1000.000 mm.	Actual Stress		129.9 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		18.000 mm.
Surface Area		13261.7 cm^2	SG of Contents		0.682
Element Volume		170200.1 cm3	Weight of Contents		112.8 Kg
Empty Weight		195.3 Kg	Operating Weight		308.2 Kg
			Comp. All. Ope. Stress		131.8 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

Required Thickness due to Internal Pressure [t_r]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (28.56 \cdot 1006.0 \cdot 0.992) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 28.56)$$

$$= 10.3502 + 3.0000 = 13.3502 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.060 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 11.0) / (0.992 \cdot 1006.0 + 0.2 \cdot 11.0)$$

$$= 30 - 0.1 = 30 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.0) / (1.0 \cdot 1000.0 + 0.2 \cdot 14.0)$$

$$= 39 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S_{act}]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (28.56 \cdot (0.992 \cdot 1006.0 + 0.2 \cdot 11.0)) / (2 \cdot 1.0 \cdot 11.0)$$

$$= 129.863 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (28.56 \cdot 503.0) / (138.0 \cdot 1.0 - 0.6 \cdot 28.56) + 3.0$$

$$= 13.542 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.060 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (503.0 + 0.6 \cdot 15.0)$$

$$= 40 - 0.1 = 40 \text{ bars}$$

Factor K, corroded condition [K_{cor}]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$



$$= (2 + (1006.0 / (2 \cdot 253.0))^2) / 6$$

$$= 0.992118$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 7.542 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 14.0, tr = 10.795, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.981$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 18.0, tr = 11.0, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.733$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	29.786 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	14.000 mm.	Basic MDMT	-35 °C
Stress/Thk. Ratio	0.981	Temperature Difference	1 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-36 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:					
Design Pressure	P	28.560 bars	Int. Design Temperature	65 °C	
Material		SA-516 70	Ext. Design Temperature	0 °C	
Length		2600.000 mm.	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1000.000 mm.	Actual Stress	132.3 N./mm^2	
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.	14.000 mm.	
Surface Area		83968.5 cm^2	SG of Contents	0.682	
Element Volume		2042401.1 cm3	Weight of Contents	1320.6 Kg	
Empty Weight		898.7 Kg	Operating Weight	2219.3 Kg	
			Comp. All. Ope. Stress	131.8 N./mm^2	

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (28.56 \cdot 503.0) / (138.0 \cdot 1.0 - 0.6 \cdot 28.56)$$

$$= 10.5418 + 3.0000 = 13.5418 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.060 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (503.0 + 0.6 \cdot 11.0)$$

$$= 30 - 0.1 = 30 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (500.0 + 0.6 \cdot 14.0)$$

$$= 38 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (28.56 \cdot (503.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0)$$

$$= 132.320 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 1.381 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 11.0, c = 3.0 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 1.0, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -18 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	29.786 bars
UCS-66(b) red.	No	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	14.000 mm.	Basic MDMT	-18 °C

PV Elite 28 U1 Licensee: SL User
FileName : 402-V-004 - Copy 7%
Element: Shell

Step: 5 5:47pm May 26,2026

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HEXAGON

Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-18 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Right D end

Elliptical Head Design Information:					
Design Pressure	P	28.560 bars	Int. Design Temperature		65 °C
Material		SA-516 70	Ext. Design Temperature		0 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	138.0 N./mm^2
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm^2
Inside Diameter	Di	1000.000 mm.	Actual Stress		129.9 N./mm^2
Spec. Min. Thk t _e , or s _a		14.0000 mm.	Spec. Nominal Thk.		18.000 mm.
Surface Area		13261.7 cm^2	SG of Contents		0.682
Element Volume		170200.1 cm3	Weight of Contents		112.8 Kg
Empty Weight		195.3 Kg	Operating Weight		308.2 Kg
			Comp. All. Ope. Stress		131.8 N./mm^2
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (28.56 \cdot 1006.0 \cdot 0.992) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 28.56)$$

$$= 10.3502 + 3.0000 = 13.3502 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.060 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 11.0) / (0.992 \cdot 1006.0 + 0.2 \cdot 11.0)$$

$$= 30 - 0.1 = 30 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 14.0) / (1.0 \cdot 1000.0 + 0.2 \cdot 14.0)$$

$$= 39 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (28.56 \cdot (0.992 \cdot 1006.0 + 0.2 \cdot 11.0)) / (2 \cdot 1.0 \cdot 11.0)$$

$$= 129.863 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (28.56 \cdot 503.0) / (138.0 \cdot 1.0 - 0.6 \cdot 28.56) + 3.0$$

$$= 13.542 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.060 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 15.0) / (503.0 + 0.6 \cdot 15.0)$$

$$= 40 - 0.1 = 40 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$



$$= (2 + (1006.0 / (2 * 253.0)))^2 / 6$$
$$= 0.992118$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 * t_{nom} / R_f) * (1 - R_f / R_o)$ 7.542 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 14.0, tr = 10.795, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.981$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 18.0, tr = 11.0, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.733$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-42 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	29.786 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-38 °C
Governing Thk.	14.000 mm.	Basic MDMT	-35 °C
Stress/Thk. Ratio	0.981	Temperature Difference	1 °C
UCS-66(c) red.	Yes	Computed Min. Temp.	-36 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 12.500 mm.
Head inner surface maximum deviation inside the specified shape, 0.625% of D: 6.250 mm.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	38.643 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	37.050 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	49.300 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	32.698 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	40.755 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	40.755 bars
App 27-4	= M.A.W.P.	29.725 bars

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 29.725 • 1.0
 = 38.643 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left D end	138.8	234.0	0.593	38.74
Shell	140.7	234.0	0.601	38.74
Right D end	138.8	234.0	0.593	38.74

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_20"	Nozzle	138.00	138.00	1.000
M1_20"	Pad	138.00	138.00	1.000
N7A_2"	Nozzle	138.00	138.00	1.000
N7B_2"	Nozzle	138.00	138.00	1.000
N3_1"	Nozzle	138.00	138.00	1.000
N1_6"	Nozzle	118.00	118.00	1.000
N1_6"	Pad	138.00	138.00	1.000
N2_6"	Nozzle	118.00	118.00	1.000
N2_6"	Pad	138.00	138.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000
N5_1"	Nozzle	138.00	138.00	1.000
N8_6"	Nozzle	118.00	118.00	1.000
N8_6"	Pad	138.00	138.00	1.000
N9_2"	Nozzle	138.00	138.00	1.000
N10_2"	Nozzle	138.00	138.00	1.000
N11_4"	Nozzle	118.00	118.00	1.000
N11_4"	Pad	138.00	138.00	1.000
N12_4"	Nozzle	118.00	118.00	1.000
N12_4"	Pad	138.00	138.00	1.000
N6_1 1/2"	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Right D end	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):



Description	Ambient	Operating	Ratio
M1_20"	84.33	179.40	0.470
N7A_2"	8.19	179.40	0.046
N7B_2"	8.19	179.40	0.046
N3_1"	8.48	179.40	0.047
N1_6"	32.40	153.40	0.211
N2_6"	32.40	153.40	0.211
N4_2"	8.19	225.00	0.036
N5_1"	8.48	179.40	0.047
N8_6"	32.40	153.40	0.211
N9_2"	8.19	225.00	0.036
N10_2"	8.19	225.00	0.036
N11_4"	21.20	153.40	0.138
N12_4"	21.20	153.40	0.138
N6_1 1/2"	9.71	179.40	0.054

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	38.643 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	37.050 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	49.300 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	32.698 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	40.755 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	40.755 bars
App 27-4	= M.A.W.P.	29.725 bars

UG-99(b), Test Pressure Calculation:

= Test Factor • MAWP • Stress Ratio

= 1.3 • 29.725 • 1.0

= 38.643 bars

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Left D end	176.2	234.0	0.753	38.74
Shell	179.5	234.0	0.767	38.74
Right D end	176.2	234.0	0.753	38.74

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
M1_20"	Nozzle	138.00	138.00	1.000
M1_20"	Pad	138.00	138.00	1.000
N7A_2"	Nozzle	138.00	138.00	1.000
N7B_2"	Nozzle	138.00	138.00	1.000
N3_1"	Nozzle	138.00	138.00	1.000
N1_6"	Nozzle	118.00	118.00	1.000
N1_6"	Pad	138.00	138.00	1.000
N2_6"	Nozzle	118.00	118.00	1.000
N2_6"	Pad	138.00	138.00	1.000
N4_2"	Nozzle	138.00	138.00	1.000
N5_1"	Nozzle	138.00	138.00	1.000
N8_6"	Nozzle	118.00	118.00	1.000
N8_6"	Pad	138.00	138.00	1.000
N9_2"	Nozzle	138.00	138.00	1.000
N10_2"	Nozzle	138.00	138.00	1.000
N11_4"	Nozzle	118.00	118.00	1.000
N11_4"	Pad	138.00	138.00	1.000
N12_4"	Nozzle	118.00	118.00	1.000
N12_4"	Pad	138.00	138.00	1.000
N6_1 1/2"	Nozzle	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Left D end	138.00	138.00	1.000
Shell	138.00	138.00	1.000
Right D end	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):



Description	Ambient	Operating	Ratio
M1_20"	84.33	179.40	0.470
N7A_2"	8.19	179.40	0.046
N7B_2"	8.19	179.40	0.046
N3_1"	8.48	179.40	0.047
N1_6"	32.40	153.40	0.211
N2_6"	32.40	153.40	0.211
N4_2"	8.19	225.00	0.036
N5_1"	8.48	179.40	0.047
N8_6"	32.40	153.40	0.211
N9_2"	8.19	225.00	0.036
N10_2"	8.19	225.00	0.036
N11_4"	21.20	153.40	0.138
N12_4"	21.20	153.40	0.138
N6_1 1/2"	9.71	179.40	0.054

Elements Suitable for Test Pressure.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	1028	11	0.0014862	96.5485
20	30	2866.67	1028	11	0.00050159	50.1477
30	40	No Calc	1028	11	0.0014862	96.5485

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	14	No Calc	...	11.4783
20	30	14	No Calc	...	7.15425
30	40	14	No Calc	...	11.4783

Minimum

7

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	2866.67	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm3	Corroded Metal Wgt. kg.	Corroded ID Volume cm3	Extra due Misc % kg.
10	20	195.345	170200	162.788	173044	13.6742
20	30	898.711	2042401	708.218	2066983	62.9097
30	40	195.345	170200	162.788	173044	13.6742
Total		1289	2382801.50	1033	2413070.75	90

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Liqd	112.84	-83.3333	50	...	Liquid 10
10	Insl	10.2945	-100	...	...	Insul.
10	Nozl	493.84	-250	...	...	M1_20"
20	Sadl	132.297	490	702.5	...	Fixed saddle
20	Sadl	132.297	2210	702.5	...	New Saddle
20	Plat	1157.48	1500	590.2	...	PLAT 1
20	Liqd	1320.57	1300	50	...	Liquid 20
20	Insl	47.3611	1300	...	...	Insul.
20	Nozl	11.0815	200	439.537	287.841	N7A_2"
20	Nozl	11.0815	200	-525.4	...	N7B_2"
20	Nozl	4.84745	160	512.7	...	N3_1"
20	Nozl	44.1477	440	574.536	...	N1_6"
20	Nozl	44.1477	2230	574.536	...	N2_6"
20	Nozl	11.576	2530	525.4	...	N4_2"
20	Nozl	4.68086	1950	512.7	...	N5_1"
20	Nozl	44.1477	1360	574.536	...	N8_6"
20	Nozl	11.0815	2500	-525.4	...	N9_2"
20	Nozl	6.78972	1060	525.4	...	N10_2"
20	Nozl	26.8777	1720	547.415	...	N11_4"
20	Nozl	26.6668	810	547.415	...	N12_4"
20	Wght	55	1450	...	...	clip Wt
20	Wght	30	810	...	...	Stilling well N
20	Wght	10	2530	...	...	Dip pipe (N4)
20	Wght	80	1300	...	...	Lifting attach.
30	Liqd	112.839	133.333	50	...	Liquid 30
30	Insl	10.2945	125	...	...	Insul
30	Nozl	6.58406	223.494	...	360	N6_1 1/2"

Total Weight of Each Detail Type:

Saddles	264.6
Platforms	1157.5
Liquid	1546.2
Insulation	68.0
Nozzles	747.6
Weights	175.0

Sum of the Detail Weights 3958.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1379.7	1379.7	1379.7	1379.7	1379.7	1379.7
Saddles	264.6	264.6	264.6	264.6	264.6	264.6
Nozzles	747.6	747.6	747.6	747.6	747.6	747.6
Wld Weights	175.0	175.0	175.0	175.0	175.0	175.0
Platforms	...	...	1157.5	1157.5	1157.5	1157.5
Insulation	...	...	68.0	68.0	68.0	68.0
Ope. Liquid	...	...	...	...	...	1546.2



Test Liquid	...	2381.3	...	...	...	...
Totals	2566.8	4948.2	3792.2	3792.2	3792.2	5338.5

Field Installation Options:

Miscellaneous Weight Percent: 7.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2566.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	4948.2 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	3792.2 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	3792.2 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	3792.2 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	5338.5 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	5065.0 kg.
Field Test Wt.	- Empty weight + test medium (full)	6173.6 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm^2	Inside Surface Area cm^2
10	20	13261.7	12410.6
20	30	83968.5	81681.4
30	40	13261.7	12410.6

Total	110491.9	106502.7 cm2
Total	118.9	114.6 ft ²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Ope.	Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
M1_20"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N7A_2"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N7B_2"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N3_1"	97.74	102.00	65	600	GR 1.1	...	...	...	...
N1_6"	48.87	51.00	65	300	GR 1.1	31.710	41.596	48.871	41.596
N2_6"	48.87	51.00	65	300	GR 1.1	31.710	41.596	48.871	41.596
N4_2"	97.74	102.00	65	600	GR 1.1	63.122	83.490	97.741	83.490
N5_1"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N8_6"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N9_2"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N10_2"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N11_4"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N12_4"	48.87	51.00	65	300	GR 1.1	...	...	...	...
N6_1 1/2"	48.87	51.00	65	300	GR 1.1	...	...	...	...
Min Rating	41.596	51.000, [for Core Elements]					...	...	...

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kg	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N1_6"	196.088	0.50	459	408	29.725	41.596	Pass
N2_6"	196.088	0.50	459	408	29.725	41.596	Pass
N4_2"	77.851	0.50	57	135	29.725	83.490	Pass

Evaluating Flange on Nozzle [N16"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(459) + 4 \cdot \text{abs}(408) 196.088 \leq \pi \cdot 196.088^3 [(48.871 - 29.725) + .500 \cdot 48.871]$$

$$= 0.76596E+04 \leq 0.10527E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N16" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.871, 0.500 \cdot 48.871 + 48.871 - 7659.6 / (\pi \cdot 196.088^3))$$

$$= \min(48.871, 41.596)$$

$$= 41.596 \text{ bars}$$

Evaluating Flange on Nozzle [N26"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(459) + 4 \cdot \text{abs}(408) 196.088 \leq \pi \cdot 196.088^3 [(48.871 - 29.725) + .500 \cdot 48.871]$$

$$= 0.76596E+04 \leq 0.10527E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N26" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(48.871, 0.500 \cdot 48.871 + 48.871 - 7659.6 / (\pi \cdot 196.088^3))$$

$$= \min(48.871, 41.596)$$

$$= 41.596 \text{ bars}$$

Evaluating Flange on Nozzle [N42"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR]$$

$$= 16(57) + 4 \cdot \text{abs}(135) 77.851 \leq \pi \cdot 77.851^3 [(97.741 - 29.725) + .500 \cdot 97.741]$$



= 0.95418E+03 ≤ 0.17669E+04 Check Passes

Maximum Pressure for Flange on Nozzle N42" [Pmax]:
= min(PR, FM • PR + PR - (16•ME + 4•FE • G) / (π • G³))
= min(97.741, 0.500 • 97.741 + 97.741 - 954.2/(π • 77.851³))
= min(97.741, 83.490)
= 83.490 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (Part 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1]	1.0
Terrain Category		2
Equipment Class		Class A
Topography Factor	[k3]	1.0
Importance Factor	[k4]	1.15
Damping Factor (Beta) for Wind (Ope)		0.0100
Damping Factor (Beta) for Wind (Empty)		0.0000
Damping Factor (Beta) for Wind (Filled)		0.0000
Use Gust Response Factor (Dynamic Analysis)		No

Calculated Values:

Vessel Length to Diameter ratio: 2.773

Design Wind Speed [Vz]:

= Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)

= 50.0 • 1.0 • 1.0 • 1.0 • 1.15

= 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:

= max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)

= max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)

= 1983.5 N/m²

Element Wind Load (operating condition):

= Wind Area • Cf • pd

= 0.3 • 0.726 • 1983.503

= 445.3 N

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	1225.00	1.0000	1.0000	1.0000	57.50	0.7258
20	1225.00	1.0000	1.0000	1.0000	57.50	0.7258
30	1225.00	1.0000	1.0000	1.0000	57.50	0.7258

Platform Load Calculations

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf	Cf
PLAT 1	51271.10	1225.00	202.35	1037.46	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm2	Elevation mm.	Pressure Kgs/m2	Force Kgf
clip Wt	0.00	1225.00	202.35	0.00
Stilling well	0.00	1225.00	202.35	0.00
Dip pipe (N4	0.00	1225.00	202.35	0.00
Lifting attac	0.00	1225.00	202.35	0.00

Wind Load Calculation:

	Wind	Wind	Wind	Wind	Element
--	------	------	------	------	---------



From	To	Height mm.	Diameter mm.	Area cm^2	Pressure Kgs/m^2	Wind Load Kgf
10	20	1225	1293.6	3093.65	202.347	49.9569
20	30	1225	1293.6	33633.6	202.347	1684.33
30	40	1225	1293.6	3093.65	202.347	49.9569

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User I 2.0000
Zone Number 2

Soil Type Rocky or Hard Soil
Response reduction factor (Force factor) R 2.0000
Time Period of Vibration (program or user) 0.0303 Seconds
Value of Sa/g from user Spectrum Data 0.0939
Percent Damping 2.0 %
Considering MCE for load determination No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
= (Sa/g)ss • DampVal) • (I / R)
= (0.0939 • 1.0) • 1.0
= 0.094

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:
= Earthquake Weight • Ah • Scalar
= 960.08 • 0.0939 • 1.3
= 117.1 Kgf

Vertical load per 8.4:
= 2/3 • Ah • W • Scalar
= 2/3 • 0.0939 • 4800 • 1.3
= 390.5 Kgf

Earthquake Load Calculation:

Earthquake	Earthquake	Element
------------	------------	---------

From	To	Height mm.	Weight Kgf	Ope Load Kgf
10	20	500	960.079	117.142
20	Sad1	500	960.079	117.142
Sad1	30	500	960.079	117.142
20	30	500	960.079	117.142
30	40	500	960.079	117.142

The Earthquake Weight does not include the weight of the saddles.

The Earthquake Loads calculated and printed in the Earthquake Load calculation report have been factored by the input scalar/load reduction factor of: 1.300.

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Shop/Field Installation Options :

Platform(s) installed in the Shop.
Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	1400.000 mm.
Center of Gravity of the Platforms	1550.000 mm.
Center of Gravity of the Liquid	1349.999 mm.
Center of Gravity of the Insulation	1346.212 mm.
Center of Gravity of the Nozzles	310.460 mm.
Center of Gravity of the Added Weights (Operating)	1383.428 mm.
Center of Gravity of the Added Weights (Empty)	1383.428 mm.
Center of Gravity of Bare Shell New and Cold	1350.000 mm.
Center of Gravity of Bare Shell Corroded	1350.000 mm.
Vessel CG in the Operating Condition	1245.993 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	1211.087 mm.
Vessel CG in the Test Condition	1264.670 mm.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(125.0 + 1.56 \cdot \sqrt{ 508.5 \cdot 11.0 }, 2 \cdot 540.0)$$

$$= 241.6719 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	508.50	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	2700.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	540.00	mm.
Saddle Width	b1 or b	125.00	mm.
Saddle Bearing Angle	delta or theta	128.00	degrees
Inside or Mean Depth of Head	Hi or hm	258.50	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	925.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		5697.40	Kgf
Pressure used in Analysis	P	28.527	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	65.38	138.00
Long. Stress at Bottom of Midspan	66.50	138.00
Long. Stress at Top of Saddles	72.49	138.00
Long. Stress at Bottom of Saddles	62.27	138.00
Tangential Shear in Shell	5.58	110.40
Circ. Stress at Horn of Saddle	53.71	172.50
Circ. Compressive Stress in Shell	1.54	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(1784.2 / 2 + 0) \cdot 925.0 / 934.7458$$

$$= 2648.5 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(933.57, 0) \cdot 925.0 / 1720.0$$

$$= 502.1 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake FI [Fsl]:

$$= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(585.71, 0) \cdot 925.0 / 1720.0$$

$$= 315.0 \text{ Kgf}$$



Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft}/\text{Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0 (586/2 + 0) \cdot 925.0/934.7458$
 $= 869.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max (\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 3049 + \max (502, 2648, 315, 869)$
 $= 5697.4 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (2.88 + \pi/4 (1.078 \cdot 1.2)^2)$
 $= 9155.089 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5829.70	Kgf
Transverse Shear Load	892.12	Kgf
Longitudinal Shear Load	933.57	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min (2 (\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\theta/2), 2 \cdot R_m)$
 $= \min (2 (1000.0/2 + 14.0 + 6.0) \sin(128.0/2), 2 \cdot 508.5)$
 $= 934.746 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1197	K2 = 1.0496	K3 = 0.7508	K4 = 0.3694
K5 = 0.7322	K6 = 0.0464	K7 = 0.0464	K8 = 0.3320
K9 = 0.2584	K10 = 0.0523	K1* = 0.2140	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -5697 \cdot 540.0 [1 - (1 - 540.0/2700.0 + (508.5^2 - 258.5^2)/(2 \cdot 540.0 \cdot 2700.0)) / (1 + (4 \cdot 258.5)/(3 \cdot 2700.0))]$
 $= -714.6 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 5697 \cdot 2700/4 ((1 + 2(509^2 - 259^2)/(2700^2)) / (1 + (4 \cdot 259)/(3 \cdot 2700)) - 4 \cdot 540/2700)$
 $= 513.2 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 28.527 \cdot 508.5/(2 \cdot 11.0) - 513.2/(\pi \cdot 508.5^2 \cdot 11.0)$
 $= 65.38 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
 $= 28.527 \cdot 508.5/(2 \cdot 11.0) + 513.2/(\pi \cdot 508.5^2 \cdot 11.0)$
 $= 66.50 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$
 $= 28.527 \cdot 508.5/(2 \cdot 11.0) - 714.6/(0.1197 \cdot \pi \cdot 508.5^2 \cdot 11.0)$
 $= 72.49 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot R_m/(2t) + M_1/(K_1^* \cdot \pi \cdot R_m^2 \cdot t)$
 $= 28.527 \cdot 508.5/(2 \cdot 11.0) + 714.6/(0.214 \cdot \pi \cdot 508.5^2 \cdot 11.0)$
 $= 62.27 \text{ N./mm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L - 2a)/(L + (4 \cdot h_m/3))$
 $= 5697 (2700.0 - 2 \cdot 540.0) / (2700.0 + (4 \cdot 258.5/3))$



= 3031.5 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.0496 \cdot 3031.46 / (508.5 \cdot 11.0)$$

$$= 5.58 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(508.5 \cdot 11.0)}$$

$$= 58.336 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7322 \cdot 5697 \cdot 0.1 / (11.0(125.0 + 58.34 + 58.34))$$

$$= -1.54 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -5697 / (4 \cdot 11.0(125.0 + 58.336 + 58.336)) -$$

$$12 \cdot 0.046 \cdot 5697 \cdot 508.5 / (2700.0 \cdot 11.0^2)$$

$$= -53.71 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(125.0 + 1.56 \cdot \sqrt{(508.5 \cdot 11.0)}, 2 \cdot 540.0)$$

$$= 241.67 \text{ mm.}$$

Distance between Saddle Supports [Ls]:

$$= 1720.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 \cdot 1720.0(65.0 - 21.1)$$

$$= 0.896 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	965.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(503.0 \cdot 11.0)} + 180.0)11.0^3)/12$$

$$= 3.284 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 3.284 + 32.564(180.781 - 5.5)^2$$

$$= 10008.206 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	296.0	11.0	5.5	32.6	17910.4	0.100E+05
Wearplate	180.0	6.0	14.0	10.8	15120.0	0.300E+04
Web	16.0	389.0	211.5	62.2	1316375.8	0.844E+04
BasePlate	150.0	16.0	414.0	24.0	993599.9	0.131E+05
Totals	...	...	...	129.6	2343006.0	0.345E+05



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 2343006/129.604 \\ &= 180.781 \text{ mm.} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 128.0/2 \\ &= 116.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\ &= (1 + \cos(116.0) - 0.5 \cdot \sin(116.0)^2) / (\pi - 2.025 + \sin(116.0) \cos(116.0)) \\ &= 0.2181 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.218 \cdot 5697.402 \\ &= 1242.8082 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 1.2560 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 147.6000 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 925.0 - 503.0 \cdot \sin(128.0/2) / 1.117 \\ &= 520.265 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 646.6024 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 3.3220 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 164.0000 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) } \\ &= \sqrt{ (3(5697 + 132)150.0 / (4 \cdot 965.0 \cdot 164.0)) } \\ &= 6.375 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 128.0/2) (500.0 + 14.0 + 6.0) \\ &= 934.746 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 934.7458 / (3 - 1) \\ &= 467.373 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 467.3729 \cdot 150.0/2 \\ &= 350.530 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 5829.699 / (965.0 \cdot 150.0) \\ &= 4.027 \text{ Kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 350.5 \cdot 4.03 \\ &= 1411.732 \text{ Kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 17.44 + 37.39 \\ &= 54.830 \text{ cm}^2 \end{aligned}$$



Compressive Stress [Sc]:

$$= P/A_r$$

$$= 1411.7/54.8298$$

$$= 2.525 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	125.0	...	20.0	...	260.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 125.0 - 16.0$$

$$= 109.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/54.83$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(260.4/54.83)}$$

$$= 21.793 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 925.0 - \cos(128.0/2) (500.0 + 14.0 + 6.0) - 16.0$$

$$= 681.047 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/246.0)}$$

$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 681.047/21.793$$

$$= 31.250$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 933.6 / (2 \cdot 965.0) \cdot 467.373 \cdot 681.05/2$$

$$= 76.986 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($31.2501 < 126.6669$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (31.25)^2 / (2 \cdot 126.67^2)) 246 / (5/3 + 3 \cdot (31.25) / (8 \cdot 126.67) - (31.25^3) / (8 \cdot 126.67^3))$$

$$= 135.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.53/135.73 + (76.99 \cdot 62.5/2604167) / 164.0$$

$$= 0.129$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	109.0	0.0	17.4	0.0	260.



Web	467.4	16.0	0.0	74.8	0.0	16.0
Totals	...	...	...	92.2	...	276.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/92.22$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 925.0 - \cos(520.0/2)(16.0 + 0.0 + 0.0) - 0.0$$

$$= 389.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(275.8/92.22)}$$

$$= 17.294 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 389.0/17.294$$

$$= 22.493$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 933.573/(2 \cdot 965.0)$$

$$= 0.484 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.484 \cdot 467.373 \cdot 389.0$$

$$= 87.945 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 87.945/44131.762$$

$$= 19.543 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (22.4929 < 126.6669) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^2 / (8 \cdot C_c^3))$$

$$= (1 - (22.49)^2 / (2 \cdot 126.67^2)) 246 /$$

$$(5/3 + 3 \cdot (22.49) / (8 \cdot 126.67) - (22.49)^2 / (8 \cdot 126.67^3))$$

$$= 139.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 3.0/139.75 + (87.95 \cdot 62.5/275.823) / 164.0$$

$$= 0.141$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	137.30 N./mm2
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	3.3800 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	1.7089 cm2
Saddle Load Q0 (Weight)	Q0	3181.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	502.1 Kg
Maximum Transverse Force	Ft	892.1 Kg
Maximum Longitudinal Force	Fl	933.6 Kg



Saddle Bolted to Steel Foundation

No

Friction force due to thermal growth, [Ff]:
= 227.4 Kg

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= max(Ff, Fl) / (Bolt Area • Number of Bolts)
= 933.6 / (1.709 • 2)
= 26.8 N./mm². Must be less than 107.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= Ft / (Bolt Area • Number of Bolts)
= 892.1 / (1.709 • 2)
= 25.6 N./mm². Must be less than 107.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= Fl / (BoltShearAllowable • Nbolts)
= 933.57 / (107.87 • 2.0)
= 0.4244 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 925.0 • 892.12 + 0.0
= 825.23 Kg-m.

Eccentricity (e):
= Rmom / QO
= 825.23 / 3181.24
= 259.40 mm. > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis
= 965.0 / 2 - 50.8
= 431.70 mm.

Modular Ratio Of Steel/Concrete (n1):
= ES / EC
= 203402.5 / 21526.32
= 9.45

K1 = 3 (e - 0.5 • Bplen)
= 3 (259.4 - 0.5 • 965.0)
= -669.30 mm.

K2 = 6 • n1 • At / Bpwid • (f + e)
= 6 • 9.45 • 1.71 / 150.0 • (431.7 + 259.4)
= 44637.12 mm.²

K3 = -K2 • (0.5 • Bplen + f)
= -44637.12 • (0.5 • 965.0 + 431.7)
= -40807252.56 mm.³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -669.3 \cdot Y^2 + 44637.12 \cdot Y + -0.4E+08 = 0$
Y = 690.28 mm.

Num = (Bplen / 2 - Y / 3 - e)
= (965.0 / 2 - 690.28 / 3 - 259.4)
= -6.99

Denom = (Bplen / 2 - Y / 3 + f)
= (965.0 / 2 - 690.28 / 3 + 431.7)
= 684.11



Total Bolt Tension Force [Tforce]:

$$\begin{aligned} &= - QO \cdot Num / Denom \\ &= - 3181.24 \cdot -6.99/684.11 \\ &= 32.52 \text{ Kgf} \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= Tforce / (Stba \cdot Nbt) \\ &= 32.52 / (137.3 \cdot 1.0) \\ &= 0.0232 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0.0, 0.4244, 0.0232] \\ &= 0.4244 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(3181.24 + 32.52) / (690.28 \cdot 150.0) \\ &= 6.09 \text{ bars} \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned} &= (Bplen - WebIngh) / 2 \\ &= (965.0 - 914.2) / 2 \\ &= 25.4000 \text{ mm.} \end{aligned}$$

Overturning Moment due To Bolt Tension [Mt]:

$$\begin{aligned} &= Tforce \cdot Adist \\ &= 32.52 \cdot 25.4 \\ &= 0.83 \text{ Kg-m.} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= fc \cdot (Y - Adist) / Y \\ &= 6.09 \cdot (690.28 - 25.4) / 690.28 \\ &= 5.86 \text{ bars} \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc) \\ &= (25.4^2 \cdot 150.0 / 6) \cdot (5.86 + 2 \cdot 6.09) \\ &= 2.97 \text{ Kg-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned} &= \sqrt{(6 \cdot \max(Mt, Mc) / (Bpwid \cdot Sba))} \\ &= \sqrt{(6 \cdot \max(0.83, 2.97) / (150.0 \cdot 207.0))} \\ &= 2.3712 \text{ mm.} \end{aligned}$$

MDMT Calculations for: Fixed saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 6.0, c = 3.0 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 1.0, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$\begin{aligned} &= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a) \\ &= \min(125.0 + 1.56 \cdot \sqrt{508.5 \cdot 11.0}, 2 \cdot 540.0) \\ &= 241.6719 \text{ mm.} \end{aligned}$$



Input and Calculated Values:

Vessel Mean Radius	Rm	508.50	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	2700.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	540.00	mm.
Saddle Width	b1 or b	125.00	mm.
Saddle Bearing Angle	delta or theta	128.00	degrees
Inside or Mean Depth of Head	Hi or hm	258.50	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	925.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		4790.38	Kgf
Pressure used in Analysis	P	28.527	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	65.47	138.00
Long. Stress at Bottom of Midspan	66.41	138.00
Long. Stress at Top of Saddles	71.45	138.00
Long. Stress at Bottom of Saddles	62.86	138.00
Tangential Shear in Shell	4.69	110.40
Circ. Stress at Horn of Saddle	45.16	172.50
Circ. Compressive Stress in Shell	1.29	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0 (1784.2/2 + 0) \cdot 925.0/934.7458$
 $= 2648.5 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max (\text{Fl,wind; Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max (933.57, 0) \cdot 925.0/1720.0$
 $= 502.1 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max (\text{Fl,eq; Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max (585.71, 0) \cdot 925.0/1720.0$
 $= 315.0 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr} (\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0 (586/2 + 0) \cdot 925.0/934.7458$
 $= 869.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max (\text{Fwl, Fwt, Fsl, Fst})$
 $= 2142 + \max (502, 2648, 315, 869)$
 $= 4790.4 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (2.88 + \pi/4 (1.078 \cdot 1.2)^2)$
 $= 9155.089 \text{ N}$



Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4922.67	Kgf
Transverse Shear Load	892.12	Kgf
Longitudinal Shear Load	933.57	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$$

$$= \min(2(1000.0/2 + 14.0 + 6.0) \sin(128.0/2), 2 \cdot 508.5)$$

$$= 934.746 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1197	K2 = 1.0496	K3 = 0.7508	K4 = 0.3694
K5 = 0.7322	K6 = 0.0464	K7 = 0.0464	K8 = 0.3320
K9 = 0.2584	K10 = 0.0523	K1* = 0.2140	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)]$$

$$= -4790 \cdot 540.0 [1 - (1 - 540.0/2700.0 + (508.5^2 - 258.5^2) / (2 \cdot 540.0 \cdot 2700.0)) / (1 + (4 \cdot 258.5^2) / (3 \cdot 2700.0))]$$

$$= -600.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \{ (1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/3L) - 4a/L \}$$

$$= 4790 \cdot 2700 / 4 \{ (1 + 2(509^2 - 259^2)/(2700^2)) / (1 + (4 \cdot 259^2) / (3 \cdot 2700)) - 4 \cdot 540 / 2700 \}$$

$$= 431.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 28.527 \cdot 508.5 / (2 \cdot 11.0) - 431.5 / (\pi \cdot 508.5^2 \cdot 11.0)$$

$$= 65.47 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 28.527 \cdot 508.5 / (2 \cdot 11.0) + 431.5 / (\pi \cdot 508.5^2 \cdot 11.0)$$

$$= 66.41 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 28.527 \cdot 508.5 / (2 \cdot 11.0) - 600.8 / (0.1197 \cdot \pi \cdot 508.5^2 \cdot 11.0)$$

$$= 71.45 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 28.527 \cdot 508.5 / (2 \cdot 11.0) + 600.8 / (0.214 \cdot \pi \cdot 508.5^2 \cdot 11.0)$$

$$= 62.86 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 4790(2700.0 - 2 \cdot 540.0) / (2700.0 + (4 \cdot 258.5/3))$$

$$= 2548.9 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.0496 \cdot 2548.85 / (508.5 \cdot 11.0)$$

$$= 4.69 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{508.5 \cdot 11.0}$$

$$= 58.336 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7322 \cdot 4790 \cdot 0.1 / (11.0(125.0 + 58.34 + 58.34))$$

$$= -1.29 \text{ N./mm}^2$$



Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) $[\sigma_7^*]$:

$$= -Q / (4 \cdot t \cdot (b + X_1 + X_2)) - 12 \cdot K_7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -4790 / (4 \cdot 11.0 \cdot (125.0 + 58.336 + 58.336)) -$$

$$12 \cdot 0.046 \cdot 4790 \cdot 508.5 / (2700.0 \cdot 11.0^2)$$

$$= -45.16 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(125.0 + 1.56 \cdot \sqrt{ 508.5 \cdot 11.0 }, 2 \cdot 540.0)$$

$$= 241.67 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	965.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	246.0	N./mm2
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ R \cdot t } + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{ 503.0 \cdot 11.0 } + 180.0)^3) / 12$$

$$= 3.284 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C_1 - Y)^2$$

$$= 3.284 + 32.564 (180.781 - 5.5)^2$$

$$= 10008.206 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	296.0	11.0	5.5	32.6	17910.4	0.100E+05
Wearplate	180.0	6.0	14.0	10.8	15120.0	0.300E+04
Web	16.0	389.0	211.5	62.2	1316375.8	0.844E+04
BasePlate	150.0	16.0	414.0	24.0	993599.9	0.131E+05
Totals	...	...	...	129.6	2343006.0	0.345E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 2343006 / 129.604$$

$$= 180.781 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 128.0 / 2$$

$$= 116.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(116.0) - 0.5 \cdot \sin(116.0)^2) / (\pi - 2.025 + \sin(116.0) \cos(116.0))$$

$$= 0.2181$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.218 \cdot 4790.377$$

$$= 1044.9536 \text{ Kg}$$

$$\text{Tension Stress, } \sigma_t = (F_h / A_s) = 1.0560 \text{ N./mm}^2$$

$$\text{Allowed Stress, } \sigma_a = 0.6 \cdot \text{Yield Str} = 147.5999 \text{ N./mm}^2$$



Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians}) \\ &= 925.0 - 503.0 \cdot \sin(128.0/2)/1.117 \\ &= 520.265 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = F_h \cdot d = 543.6635 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } S_b &= (M \cdot C_1 / I) = 2.7931 \text{ N./mm}^2 \\ \text{Allowed Stress, } S_a &= 2/3 \cdot \text{Yield Str} = 163.9998 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))} \\ &= \sqrt{(3(4790 + 132)150.0 / (4 \cdot 965.0 \cdot 164.0))} \\ &= 5.858 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2)(\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 128.0/2)(500.0 + 14.0 + 6.0) \\ &= 934.746 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 934.7458 / (3 - 1) \\ &= 467.373 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 467.3729 \cdot 150.0 / 2 \\ &= 350.530 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 4922.674 / (965.0 \cdot 150.0) \\ &= 3.401 \text{ Kg/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= A_p \cdot B_p \\ &= 350.5 \cdot 3.4 \\ &= 1192.085 \text{ Kg} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 17.44 + 37.39 \\ &= 54.830 \text{ cm}^2 \end{aligned}$$

Compressive Stress [Sc]:

$$\begin{aligned} &= P / A_r \\ &= 1192.1 / 54.8298 \\ &= 2.132 \text{ N./mm}^2 \end{aligned}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	125.0	...	20.0	...	260.

Rib dimension [D]:

$$\begin{aligned} &= \text{Saddle Width} - \text{Web Thickness} \\ &= 125.0 - 16.0 \\ &= 109.000 \text{ mm.} \end{aligned}$$

Distance to Centroid from Datum [ytot]:

$$\begin{aligned} &= AY / A \\ &= 0.0 / 54.83 \\ &= 0.000 \text{ mm.} \end{aligned}$$



Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{260.4/54.83}$$

$$= 21.793 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 925.0 - \cos(128.0/2)(500.0 + 14.0 + 6.0) - 16.0$$

$$= 681.047 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/246.0)}$$

$$= 126.667$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 681.047/21.793$$

$$= 31.250$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 933.6 / (2 \cdot 965.0) \cdot 467.373 \cdot 681.05/2$$

$$= 76.986 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($31.2501 < 126.6669$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (31.25)^2 / (2 \cdot 126.67^2)) 246 /$$

$$(5/3 + 3 \cdot (31.25) / (8 \cdot 126.67) - (31.25^3) / (8 \cdot 126.67^3))$$

$$= 135.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.13/135.73 + (76.99 \cdot 62.5/2604167) / 164.0$$

$$= 0.126$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	109.0	0.0	17.4	0.0	260.
Web	467.4	16.0	0.0	74.8	0.0	16.0
Totals	...	...	...	92.2	...	276.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/92.22$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 925.0 - \cos(520.0/2)(16.0 + 0.0 + 0.0) - 0.0$$

$$= 389.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{275.8/92.22}$$

$$= 17.294 \text{ mm.}$$



Slenderness ratio $[KL/r]$:

$$= KL/r$$

$$= 1 \cdot 389.0/17.294$$

$$= 22.493$$

Unit Force $[Force,u]$:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 933.573/(2 \cdot 965.0)$$

$$= 0.484 \text{ Kg/mm.}$$

Moment at base of inner Rib $[Mbase,c]$:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.484 \cdot 467.373 \cdot 389.0$$

$$= 87.945 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load $[\sigma B,base,c]$:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 87.945/44131.762$$

$$= 19.543 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (22.4929 < 126.6669) per AISC E2-1 $[Sca]$:

$$= (1-(KL/r)^2/(2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1-(22.49)^2 / (2 \cdot 126.67^2)) 246 /$$

$$(5/3 + 3 \cdot (22.49) / (8 \cdot 126.67) - (22.49^3) / (8 \cdot 126.67^3))$$

$$= 139.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 2.54/139.75 + (87.95 \cdot 62.5/275.823) / 164.0$$

$$= 0.137$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	137.30 N./mm2
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	3.3800 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BltArea	1.7089 cm2
Saddle Load Q0 (Weight)	Q0	2274.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	502.1 Kg
Maximum Transverse Force	Ft	892.1 Kg
Maximum Longitudinal Force	Fl	933.6 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction $[rb,l]$:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 933.6/(1.709 \cdot 2)$$

$$= 26.8 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction $[rb,t]$:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 892.1/(1.709 \cdot 2)$$

$$= 25.6 \text{ N./mm}^2. \text{ Must be less than } 107.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load $[Bltarearl]$:

$$= 0.0 \text{ (} Q0 > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load $[Bltarears]$:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 933.57/(107.87 \cdot 2.0)$$

$$= 0.4244 \text{ cm}^2$$

Bolt Area due to Transverse Load:



Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} &= B \cdot Ft + \text{Sum of X Moments} \\ &= 925.0 \cdot 892.12 + 0.0 \\ &= 825.23 \text{ Kg-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= Rmom / QO \\ &= 825.23/2274.22 \\ &= 362.85 \text{ mm.} > Bplen/6 \rightarrow \text{Uplift in Transverse direction} \end{aligned}$$

f = Bplen / 2 - Edgedis

$$\begin{aligned} &= 965.0/2 - 50.8 \\ &= 431.70 \text{ mm.} \end{aligned}$$

Modular Ratio Of Steel/Concrete (n1):

$$\begin{aligned} &= ES / EC \\ &= 203402.5/21526.32 \\ &= 9.45 \end{aligned}$$

K1 = 3 (e - 0.5 • Bplen)

$$\begin{aligned} &= 3 (362.85 - 0.5 \cdot 965.0) \\ &= -358.94 \text{ mm.} \end{aligned}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$\begin{aligned} &= 6 \cdot 9.45 \cdot 1.71/150.0 \cdot (431.7 + 362.85) \\ &= 51319.19 \text{ mm.}^2 \end{aligned}$$

K3 = -K2 • (0.5 • Bplen + f)

$$\begin{aligned} &= -51319.19 \cdot (0.5 \cdot 965.0 + 431.7) \\ &= -46916001.61 \text{ mm.}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + -358.94 \cdot Y^2 + 51319.19 \cdot Y + -0.5E+08 &= 0 \\ Y &= 465.32 \text{ mm.} \end{aligned}$$

Num = (Bplen / 2 - Y / 3 - e)

$$\begin{aligned} &= (965.0/2 - 465.32/3 - 362.85) \\ &= -35.46 \end{aligned}$$

Denom = (Bplen / 2 - Y / 3 + f)

$$\begin{aligned} &= (965.0/2 - 465.32/3 + 431.7) \\ &= 759.09 \end{aligned}$$

Total Bolt Tension Force [Tforce]:

$$\begin{aligned} &= - QO \cdot Num / Denom \\ &= - 2274.22 \cdot -35.46/759.09 \\ &= 106.25 \text{ Kgf} \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= Tforce / (Stba \cdot Nbt) \\ &= 106.25/(137.3 \cdot 1.0) \\ &= 0.0759 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0.0, 0.4244, 0.0759] \\ &= 0.4244 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(2274.22 + 106.25)/(465.32 \cdot 150.0) \\ &= 6.69 \text{ bars} \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$= (Bplen - WebIngh) / 2$$



$$= (965.0 - 914.2)/2 \\ = 25.4000 \text{ mm.}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{\text{force}} \cdot A_{\text{dist}} \\ = 106.25 \cdot 25.4 \\ = 2.70 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= f_c \cdot (Y - A_{\text{dist}}) / Y \\ = 6.69 \cdot (465.32 - 25.4) / 465.32 \\ = 6.32 \text{ bars}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (A_{\text{dist}}^2 \cdot B_{\text{pwid}} / 6) \cdot (f1 + 2 \cdot f_c) \\ = (25.4^2 \cdot 150.0 / 6) \cdot (6.32 + 2 \cdot 6.69) \\ = 3.24 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(Mt, Mc) / (B_{\text{pwid}} \cdot S_{ba}))} \\ = \sqrt{(6 \cdot \max(2.7, 3.24) / (150.0 \cdot 207.0))} \\ = 2.4781 \text{ mm.}$$

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govern. thk, tg = 6.0, c = 3.0 mm., E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 1.0, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \cdot \sqrt{ R_m \cdot t }, 2a)$$

$$= \min(125.0 + 1.56 \cdot \sqrt{ 507.0 \cdot 14.0 }, 2 \cdot 540.0)$$

$$= 256.4294 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	507.00	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	2700.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	540.00	mm.
Saddle Width	b1 or b	125.00	mm.
Saddle Bearing Angle	delta or theta	128.00	degrees
Inside or Mean Depth of Head	Hi or hm	257.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	925.00	mm.
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		4225.04	Kgf
Pressure used in Analysis	P	38.693	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	69.73	234.00
Long. Stress at Bottom of Midspan	70.40	234.00
Long. Stress at Top of Saddles	73.90	234.00
Long. Stress at Bottom of Saddles	67.92	234.00
Tangential Shear in Shell	3.26	138.58
Circ. Stress at Horn of Saddle	25.00	350.99
Circ. Compressive Stress in Shell	0.85	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr}(F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0(588.8 / 2 + 0) \cdot 925.0 / 934.7458$$

$$= 874.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(308.08, 0) \cdot 925.0 / 1720.0$$

$$= 165.7 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 3351 + \max(166, 874, 0, 0)$$

$$= 4225.0 \text{ Kgf}$$



Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(2.88 + \pi/4(1.078 \cdot 1.2)^2)$$

$$= 9155.089 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4357.33	Kgf
Transverse Shear Load	294.40	Kgf
Longitudinal Shear Load	308.08	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(1000.0/2 + 14.0 + 6.0)\sin(128.0/2), 2 \cdot 507.0)$$

$$= 934.746 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1197	K2 = 1.0496	K3 = 0.7508	K4 = 0.3694
K5 = 0.7322	K6 = 0.0464	K7 = 0.0464	K8 = 0.3320
K9 = 0.2584	K10 = 0.0523	K1* = 0.2140	

Note: Dimension a is greater than or equal to R_m/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -4225 \cdot 540.0 [1 - (1 - 540.0/2700.0 + (507.0^2 - 257.0^2)/(2 \cdot 540.0 \cdot 2700.0))/(1 + (4 \cdot 257.0)/(3 \cdot 2700.0))]$$

$$= -529.3 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 4225 \cdot 2700/4((1 + 2(507^2 - 257^2)/(2700^2))/(1 + (4 \cdot 257)/(3 \cdot 2700)) - 4 \cdot 540/2700)$$

$$= 381.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 38.693 \cdot 507.0/(2 \cdot 14.0) - 381.8/(\pi \cdot 507.0^2 \cdot 14.0)$$

$$= 69.73 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 38.693 \cdot 507.0/(2 \cdot 14.0) + 381.8/(\pi \cdot 507.0^2 \cdot 14.0)$$

$$= 70.40 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 38.693 \cdot 507.0/(2 \cdot 14.0) - 529.3/(0.1197 \cdot \pi \cdot 507.0^2 \cdot 14.0)$$

$$= 73.90 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$= P \cdot R_m/(2t) + M1/(K1* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 38.693 \cdot 507.0/(2 \cdot 14.0) + 529.3/(0.214 \cdot \pi \cdot 507.0^2 \cdot 14.0)$$

$$= 67.92 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 4225(2700.0 - 2 \cdot 540.0)/(2700.0 + (4 \cdot 257.0/3))$$

$$= 2249.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ₂]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.0496 \cdot 2249.53/(507.0 \cdot 14.0)$$

$$= 3.26 \text{ N./mm}^2$$

Decay Length (4.15.20) [x₁, x₂]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(507.0 \cdot 14.0)}$$



= 65.715 mm.

Circumferential Stress in shell, no rings (4.15.21) $[\sigma_6]$:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7322 \cdot 4225 \cdot 0.1 / (14.0(125.0 + 65.71 + 65.71))$$

$$= -0.85 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) $[\sigma_7^*]$:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -4225 / (4 \cdot 14.0(125.0 + 65.715 + 65.715)) -$$

$$12 \cdot 0.046 \cdot 4225 \cdot 507.0 / (2700.0 \cdot 14.0^2)$$

$$= -25.00 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) $[B1]$:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(125.0 + 1.56 \cdot \sqrt{(507.0 \cdot 14.0)}, 2 \cdot 540.0)$$

$$= 256.43 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	965.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell $[ShellInertia]$:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$$

$$= ((1.56 \cdot \sqrt{(500.0 \cdot 14.0)} + 180.0)14.0^3)/12$$

$$= 7.101 \text{ cm}^4$$

Resolved Inertia for the Shell $[I,shell]$:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.101 + 43.473(169.71 - 7.0)^2$$

$$= 11516.262 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	310.5	14.0	7.0	43.5	30430.9	0.115E+05
Wearplate	180.0	6.0	17.0	10.8	18360.0	0.252E+04
Web	16.0	389.0	214.5	62.2	1335047.8	0.910E+04
BasePlate	150.0	16.0	417.0	24.0	1000799.9	0.147E+05
Totals	...	...	...	140.5	2384638.5	0.378E+05

Distance to Centroid $[C1]$:

$$= AY / A$$

$$= 2384639 / 140.513$$

$$= 169.710 \text{ mm.}$$

Angle $[\beta]$:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 128.0/2$$

$$= 116.0$$

Saddle Splitting Coefficient $[K1]$:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta))$$

$$= (1 + \cos(116.0) - 0.5 \cdot \sin(116.0)^2) / (\pi - 2.025 + \sin(116.0)\cos(116.0))$$

$$= 0.2181$$

Saddle Splitting Force $[Fh]$:

$$= K1 \cdot Q$$

$$= 0.218 \cdot 4225.038$$

$$= 921.6327 \text{ Kgf}$$



Tension Stress, $St = (Fh/As) = 0.9314 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 925.0 - 500.0 \cdot \sin(128.0/2) / 1.117$
 $= 522.679 \text{ mm.}$

Bending Moment, $M = Fh * d = 481.7274 \text{ Kg-m.}$

Bending Stress, $Sb = (M * C1 / I) = 2.1202 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(4225 + 132)150.0 / (4 * 965.0 * 173.33))}$
 $= 5.361 \text{ mm.}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 128.0/2) (500.0 + 14.0 + 6.0)$
 $= 934.746 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 934.7458 / (3 - 1)$
 $= 467.373 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 467.3729 * 150.0 / 2$
 $= 350.530 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 4357.334 / (965.0 * 150.0)$
 $= 3.010 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 350.5 * 3.01$
 $= 1055.181 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 17.44 + 37.39$
 $= 54.830 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P/Ar$
 $= 1055.2 / 54.8298$
 $= 1.887 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	125.0	...	20.0	...	260.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 125.0 - 16.0$
 $= 109.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:



$$= AY / A$$

$$= 0.0/54.83$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(260.4/54.83)}$$

$$= 21.793 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 925.0 - \cos(128.0/2) (500.0 + 14.0 + 6.0) - 16.0$$

$$= 681.047 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 681.047/21.793$$

$$= 31.250$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 308.1 / (2 \cdot 965.0) \cdot 467.373 \cdot 681.05/2$$

$$= 25.405 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($31.2501 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (31.25)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (31.25) / (8 \cdot 123.21) - (31.25^3) / (8 \cdot 123.21^3))$$

$$= 143 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.89/142.99 + (25.41 \cdot 62.5/2604167) / 173.33$$

$$= 0.048$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	109.0	0.0	17.4	0.0	260.
Web	467.4	16.0	0.0	74.8	0.0	16.0
Totals	...	...	...	92.2	...	276.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/92.22$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 925.0 - \cos(520.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 389.000 \text{ mm.}$$

Radius of Gyration [r]:



$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{275.8/92.22}$$

$$= 17.294 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 389.0/17.294$$

$$= 22.493$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 308.079/(2 \cdot 965.0)$$

$$= 0.160 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.16 \cdot 467.373 \cdot 389.0$$

$$= 29.022 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 29.022/44131.762$$

$$= 6.449 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($22.4929 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.49)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (22.49) / (8 \cdot 123.21) - (22.49)^3 / (8 \cdot 123.21^3))$$

$$= 147.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (M_{base,c} \cdot C_1/I) / S_{ba}$$

$$= 2.25/147.41 + (29.02 \cdot 62.5/275.823) / 173.33$$

$$= 0.052$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	108.00 N./mm ²
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	3.3800 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.7089 cm ²
Saddle Load QO (Weight)	QO	3483.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	165.7 Kg
Maximum Transverse Force	Ft	294.4 Kg
Maximum Longitudinal Force	Fl	308.1 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 308.1 / (1.709 \cdot 2)$$

$$= 8.8 \text{ N./mm}^2. \text{ Must be less than } 64.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 294.4 / (1.709 \cdot 2)$$

$$= 8.4 \text{ N./mm}^2. \text{ Must be less than } 64.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 308.08 / (64.8 \cdot 2.0)$$



= 0.2331 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 925.0 • 294.4 + 0.0

= 272.33 Kg-m.

Eccentricity (e):

= Rmom / QO

= 272.33/3483.34

= 78.18 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 0.2331, 0.0]

= 0.2331 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:

Wear Pad Width (180.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

= min(b + $1.56 \cdot \sqrt{Rm \cdot t}$), 2a)

= min(125.0 + $1.56 \cdot \sqrt{507.0 \cdot 14.0}$), 2 • 540.0)

= 256.4294 mm.

Input and Calculated Values:

Vessel Mean Radius	Rm	507.00	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	2700.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	540.00	mm.
Saddle Width	b1 or b	125.00	mm.
Saddle Bearing Angle	delta or theta	128.00	degrees
Inside or Mean Depth of Head	Hi or hm	257.00	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	925.00	mm.
Coefficient of Friction	mu	0.10	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		3256.93	Kgf
Pressure used in Analysis	P	38.693	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		69.81	234.00
Long. Stress at Bottom of Midspan		70.32	234.00
Long. Stress at Top of Saddles		73.02	234.00
Long. Stress at Bottom of Saddles		68.41	234.00
Tangential Shear in Shell		2.51	138.58
Circ. Stress at Horn of Saddle		19.27	350.99
Circ. Compressive Stress in Shell		0.65	234.00



Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
= $F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
= $3.0(588.8/2 + 0) \cdot 925.0/934.7458$
= 874.0 Kgf

Longitudinal Saddle Reaction Force [Fwl]:
= $\max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
= $\max(308.08, 0) \cdot 925.0/1720.0$
= 165.7 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:
= Saddle Load + $\max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
= $2383 + \max(166, 874, 0, 0)$
= 3256.9 Kgf

Longitudinal Wind Force [Fl]:
= $\text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
= $1.1 \cdot 1984.306(2.88 + \pi/4(1.078 \cdot 1.2)^2)$
= 9155.089 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	3389.23	Kgf
Transverse Shear Load	294.40	Kgf
Longitudinal Shear Load	308.08	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
= $\min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
= $\min(2(1000.0/2 + 14.0 + 6.0)\sin(128.0/2), 2 \cdot 507.0)$
= 934.746 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1197	K2 = 1.0496	K3 = 0.7508	K4 = 0.3694
K5 = 0.7322	K6 = 0.0464	K7 = 0.0464	K8 = 0.3320
K9 = 0.2584	K10 = 0.0523	K1* = 0.2140	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
= $-Q \cdot a[1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$
= $-3257 \cdot 540.0[1 - (1 - 540.0/2700.0 + (507.0^2 - 257.0^2)/(2 \cdot 540.0 \cdot 2700.0))/(1 + (4 \cdot 257.0)/(3 \cdot 2700.0))]$
= -408.0 Kg-m.

Moment per Equation 4.15.2 [M2]:
= $Q \cdot L/4((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$
= $3257 \cdot 2700/4((1 + 2(507^2 - 257^2)/(2700^2))/(1 + (4 \cdot 257)/(3 \cdot 2700)) - 4 \cdot 540/2700)$
= 294.3 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
= $P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
= $38.693 \cdot 507.0/(2 \cdot 14.0) - 294.3/(\pi \cdot 507.0^2 \cdot 14.0)$
= 69.81 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
= $P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$
= $38.693 \cdot 507.0/(2 \cdot 14.0) + 294.3/(\pi \cdot 507.0^2 \cdot 14.0)$
= 70.32 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:
= $P \cdot R_m/(2t) - M_1/(K_1 \cdot \pi \cdot R_m^2 t)$
= $38.693 \cdot 507.0/(2 \cdot 14.0) - 408.0/(0.1197 \cdot \pi \cdot 507.0^2 \cdot 14.0)$
= 73.02 N./mm²



Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 38.693 \cdot 507.0 / (2 \cdot 14.0) + 408.0 / (0.214 \cdot \pi \cdot 507.0^2 \cdot 14.0)$$

$$= 68.41 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 3257(2700.0 - 2 \cdot 540.0) / (2700.0 + (4 \cdot 257.0 / 3))$$

$$= 1734.1 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.0496 \cdot 1734.08 / (507.0 \cdot 14.0)$$

$$= 2.51 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(507.0 \cdot 14.0)}$$

$$= 65.715 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7322 \cdot 3257 \cdot 0.1 / (14.0(125.0 + 65.71 + 65.71))$$

$$= -0.65 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.23) [σ_7^*]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 12 \cdot K7 \cdot Q \cdot R_m / (L \cdot t^2)$$

$$= -3257 / (4 \cdot 14.0(125.0 + 65.715 + 65.715)) -$$

$$12 \cdot 0.046 \cdot 3257 \cdot 507.0 / (2700.0 \cdot 14.0^2)$$

$$= -19.27 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(125.0 + 1.56 \cdot \sqrt{(507.0 \cdot 14.0)}, 2 \cdot 540.0)$$

$$= 256.43 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	965.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.100	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(500.0 \cdot 14.0)} + 180.0)^3) / 12$$

$$= 7.101 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 7.101 + 43.473(169.71 - 7.0)^2$$

$$= 11516.262 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	310.5	14.0	7.0	43.5	30430.9	0.115E+05
Wearplate	180.0	6.0	17.0	10.8	18360.0	0.252E+04
Web	16.0	389.0	214.5	62.2	1335047.8	0.910E+04
BasePlate	150.0	16.0	417.0	24.0	1000799.9	0.147E+05
Totals	...	...	...	140.5	2384638.5	0.378E+05



Distance to Centroid [C1]:

$$\begin{aligned} &= AY / A \\ &= 2384639/140.513 \\ &= 169.710 \text{ mm.} \end{aligned}$$

Angle [β]:

$$\begin{aligned} &= 180 - \text{Saddle Angle}/2 \\ &= 180 - 128.0/2 \\ &= 116.0 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned} &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\ &= (1 + \cos(116.0) - 0.5 \cdot \sin(116.0)^2) / (\pi - 2.025 + \sin(116.0)\cos(116.0)) \\ &= 0.2181 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned} &= K1 \cdot Q \\ &= 0.218 \cdot 3256.934 \\ &= 710.4543 \text{ Kgf} \end{aligned}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 0.7180 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned} &= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ &= 925.0 - 500.0 \cdot \sin(128.0/2) / 1.117 \\ &= 522.679 \text{ mm.} \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 371.3468 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 1.6344 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{ (3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})) } \\ &= \sqrt{ (3(3257 + 132)150.0 / (4 \cdot 965.0 \cdot 173.33)) } \\ &= 4.728 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned} &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\ &= 2 \cdot \cos(90 - 128.0/2) (500.0 + 14.0 + 6.0) \\ &= 934.746 \text{ mm.} \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned} &= \text{Web Length} / (\text{Nr ribs} - 1) \\ &= 934.7458 / (3 - 1) \\ &= 467.373 \text{ mm.} \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned} &= e \cdot \text{Bpwid} / 2 \\ &= 467.3729 \cdot 150.0/2 \\ &= 350.530 \text{ cm}^2 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned} &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\ &= 3389.23 / (965.0 \cdot 150.0) \\ &= 2.341 \text{ Kgf/cm}^2 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned} &= Ap \cdot Bp \\ &= 350.5 \cdot 2.34 \\ &= 820.743 \text{ Kgf} \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned} &= \text{Rib Area} + \text{Web Area} \\ &= 17.44 + 37.39 \\ &= 54.830 \text{ cm}^2 \end{aligned}$$



Compressive Stress [Sc]:

$$= P/A_r$$

$$= 820.7/54.8298$$

$$= 1.468 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	125.0	...	20.0	...	260.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 125.0 - 16.0$$

$$= 109.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/54.83$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(260.4/54.83)}$$

$$= 21.793 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 925.0 - \cos(128.0/2)(500.0 + 14.0 + 6.0) - 16.0$$

$$= 681.047 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 681.047/21.793$$

$$= 31.250$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 308.1/(2 \cdot 965.0) \cdot 467.373 \cdot 681.05/2$$

$$= 25.405 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($31.2501 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (31.25)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (31.25) / (8 \cdot 123.21) - (31.25^3) / (8 \cdot 123.21^3))$$

$$= 143 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.47/142.99 + (25.41 \cdot 62.5/2604167) / 173.33$$

$$= 0.045$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	109.0	0.0	17.4	0.0	260.



Web	467.4	16.0	0.0	74.8	0.0	16.0
Totals	...	...	...	92.2	...	276.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/92.22$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 125.0/2$$

$$= 62.500 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 925.0 - \cos(520.0/2)(16.0 + 0.0 + 0.0) - 0.0$$

$$= 389.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(275.8/92.22)}$$

$$= 17.294 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 389.0/17.294$$

$$= 22.493$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 308.079/(2 \cdot 965.0)$$

$$= 0.160 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.16 \cdot 467.373 \cdot 389.0$$

$$= 29.022 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 29.022/44131.762$$

$$= 6.449 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (22.4929 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (22.49)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (22.49) / (8 \cdot 123.21) - (22.49^3) / (8 \cdot 123.21^3))$$

$$= 147.4 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 1.75/147.41 + (29.02 \cdot 62.5/275.823) / 173.33$$

$$= 0.049$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-285 C
Bolt Allowable Stress	Stba	108.00 N./mm2
Bolt Corrosion Allowance	Bca	3.0 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	3.3800 cm2
BasePlate Allowable Stress	S	138.00 N./mm2
Area Available in a Single Bolt	BlArea	1.7089 cm2
Saddle Load Q0 (Weight)	Q0	2515.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	165.7 Kg
Maximum Transverse Force	Ft	294.4 Kg
Maximum Longitudinal Force	Fl	308.1 Kg
Saddle Bolted to Steel Foundation	No	



Shear Stress in a Single Bolt, Longitudinal Direction [τ_b, l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $308.1 / (1.709 \cdot 2)$
= 8.8 N./mm^2 . Must be less than 64.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [τ_b, t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $294.4 / (1.709 \cdot 2)$
= 8.4 N./mm^2 . Must be less than 64.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [$Bltarearl$]:
= 0.0 ($QO > QL \rightarrow$ No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [$Bltarears$]:
= $F_l / (\text{BoltShearAllowable} \cdot Nbolts)$
= $308.08 / (64.8 \cdot 2.0)$
= 0.2331 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [$Rmom$]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $925.0 \cdot 294.4 + 0.0$
= 272.33 Kg-m .

Eccentricity (e):
= $Rmom / QO$
= $272.33 / 2515.24$
= 108.27 mm . $< Bplen/6 \rightarrow$ No Uplift in Transverse direction

Bolt Area due to Transverse Load [$Bltareart$]:
= 0 (No Uplift)

Required Area of a Single Bolt [$Bltarear$]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.0, 0.2331, 0.0]$
= 0.2331 cm^2

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: M120"

From: 10

Pressure for Reinforcement Calculations	P	28.527	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	14.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	0.0000	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-516 70	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	140.4720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	900.0000	mm.
Thickness of Pad	te	14.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm.
Reinforcing Pad Width		184.0000	mm.
This is a Manway, Coupling or Access Opening.			
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: M120"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	20.000	in.
Actual Thickness Used in Calculation	0.472	in.



Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 28.53 \cdot 1006.0 \cdot 0.992 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 28.53)$
 $= 10.3380$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.53 \cdot 257.0 / (138 \cdot 1.0 - 0.6 \cdot 28.53)$
 $= 5.3796$ mm.

The diameter limit is greater than the head diameter.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1028.0000 mm.
 Parallel to Vessel Wall, opening length d 514.0000 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	53.137	NA	NA
Area in Shell	A1	3.403	NA	NA
Area in Nozzle Wall	A2	1.991	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.440	NA	NA
Area in Element	A5	51.520	NA	NA
TOTAL AREA AVAILABLE	Atot	59.354	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 855.5962 14.0000 mm.
 Based on given Pad Diameter: 900.0000 12.3107 mm.
 Based on Shell or Nozzle Thickness: 909.5289 12.0000 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (514.0 \cdot 10.338 \cdot 1.0 + 2 \cdot 9.0 \cdot 10.338 \cdot 1.0 \cdot (1 - 1.0))$
 $= 53.137$ cm²



Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 514.0(1.0 \cdot 11.0 - 1.0 \cdot 10.338) - 2 \cdot 9.0 \\ &\quad (1.0 \cdot 11.0 - 1.0 \cdot 10.338) \cdot (1 - 1.0) \\ &= 3.403 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (9.0 - 5.38) \cdot 1.0 \\ &= 1.991 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0 \\ &= 2.440 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (900.0 - 532.0) \cdot 14.0 \cdot 1.0 \\ &= 51.520 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 8.3796 \text{ mm.}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 8.3796 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 12.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 5.38$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.598$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 5.38$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.598$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.0$, $t_r = 5.38$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.598$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 10.338$, $c = 3.0 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.94$, Temp. Reduction = 3 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-38 °C



Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 12.0, tr = 5.38, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.598$, Temp. Reduction = 23 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-40 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Gov. MDMT of the Nozzle	:	-48 °C
Gov. MDMT of the Reinforcement Pad	:	-38 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-38 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-48 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $28.53/51.00 = 0.559$

Weld Size Calculations, Description: M120"

Intermediate Calc. for nozzle/shell Welds	Tmin	9.0000 mm.
Intermediate Calc. for pad/shell Welds	TminPad	14.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$7.0000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (53.1373 - 3.4027 + 2 \cdot 9.0 \cdot 1.0 \cdot (1.0 \cdot 11.0 - 10.338)) \cdot 138)$
 $= 70152.16 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - C_{an}/7.07)^2 \cdot f_{r2}) \cdot S_v$
 $= (1.9912 + 51.52 + 2.44 - 0.0 \cdot 1.0) \cdot 138$
 $= 78732.14 \text{ Kg}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1.9912 + 0.0 + 1. + (1.98)) \cdot 138$
 $= 6995.27 \text{ Kg}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1.9912 + 0.0 + 2.44 + 51.52 + (1.98)) \cdot 138$
 $= 81518.31 \text{ Kg}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 532.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 57621. \text{ Kg}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 900.0 \cdot 12.0 \cdot 0.49 \cdot 138$
 $= 116972. \text{ Kg}$

Shear, Nozzle Wall [Snw]:



$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 261.5) \cdot (12.0 - 3.0) \cdot 0.7 \cdot 138 \\ &= 72830. \text{ Kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 532.0 \cdot 14.0 \cdot 0.74 \cdot 138 \\ &= 121826. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 532.0 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 95721. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (116972 + 72830) = 189802 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (57621 + 121826 + 95721 + 0) = 275168 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (116972 + 95721 + 0) = 212693 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 189802 Kgf, must exceed W = 70152 Kgf	or W1 = 78732 Kgf
Path 2-2 = 275167 Kgf, must exceed W = 70152 Kgf	or W2 = 6995 Kgf
Path 3-3 = 212692 Kgf, must exceed W = 70152 Kgf	or W3 = 81518 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

The Drop for this Nozzle is : 39.6901 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 194.1621 mm.

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.308 %

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Input, Nozzle Desc: N7A2"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Cylinder/Cone Centerline	L1	430.0000	mm.
Distance from Bottom/Left Tangent		250.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		56.78	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: N7A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 = 10.5192 mm.



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 0.5939$ mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 216.6222 mm.
 Parallel to Vessel Wall, opening length d 108.3111 mm.
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 27.5000 mm.

Results of Nozzle Reinforcement Area Calculations: (cm2)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	5.697	NA	NA
Area in Shell	A1	6.217	NA	NA
Area in Nozzle Wall	A2	11.606	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
TOTAL AREA AVAILABLE	Atot	18.823	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 31.63 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (108.3111 \cdot 10.5192 \cdot 0.5 + 2 \cdot 13.764 \cdot 10.5192 \cdot 0.5 \cdot (1 - 1.0))$
 $= 5.697$ cm²

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 108.311(1.0 \cdot 11.0 - 0.5 \cdot 10.519) - 2 \cdot 13.764$
 $(1.0 \cdot 11.0 - 0.5 \cdot 10.5192) \cdot (1 - 1.0)$
 $= 6.217$ cm²

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2/\sin(\alpha3)$
 $= (2 \cdot 27.5)(13.76 - 0.59)1.0/\sin(38.6)$
 $= 11.606$ cm²

See ASME VIII-1 2011(a) Appendix L, L-7.7.7(b) for more information.

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 1.000$ cm²

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit -35 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N7A2"



Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (5.6967 - 6.2175 + 2 \cdot 13.764 \cdot 1.0 \cdot (1.0 \cdot 11.0 - 5.2596)) \cdot 138)$
 $= 1490.81 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (11.6059 + 0.0 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 17738.46 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (11.6059 + 0.0 + 1. + (3.0281)) \cdot 138$
 $= 21999.45 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (11.6059 + 0.0 + 1. + 0.0 + (3.0281)) \cdot 138$
 $= 21999.45 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 160.8038 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 17416. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 67.2787) \cdot (16.764 - 3.0) \cdot 0.7 \cdot 138$
 $= 28656. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 160.8038 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$
 $= 28933. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (17416 + 28656) = 46073 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (17416 + 0 + 28933 + 0) = 46349 \text{ Kgf}$

Failure Path [Path 3-3]:
 $= (\text{Sonw} + \text{Tngw} + \text{Sinw})$
 $= (17416 + 28933 + 0) = 46349 \text{ Kgf}$

Summary of Failure Path Calculations:

Path 1-1 = 46072	Kgf, must exceed W = 1490 Kgf	or W1 = 17738 Kgf
Path 2-2 = 46349	Kgf, must exceed W = 1490 Kgf	or W2 = 21999 Kgf
Path 3-3 = 46349	Kgf, must exceed W = 1490 Kgf	or W3 = 21999 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:



Converged Maximum Allowable Pressure in the Operating case: 30 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: N7A2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 2.000 in.
Actual Thickness Used in Calculation 0.660 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
= 10.5192 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.5 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
= 0.5939 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N7A2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5939	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.5192	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5192	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[7.8, max(13.5192, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.5939, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Weld Size Calculations, Description: N7A2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 90.6442 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 323.4677 mm.

PV Elite 28 U1 Licensee: SL User
FileName : 402-V-004 - Copy 7%
Nozzle Calcs.: N7A_2"

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Input, Nozzle Desc: N7B2"

From: 20

Pressure for Reinforcement Calculations	P	28.560	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		250.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	480.4260	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N7B2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.56 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.56)$
 $= 10.5417$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.56 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 28.56)$



= 0.5952 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N7B2".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5952	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.5417	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5417	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.8, max(13.5417, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.5952, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.542, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.958$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N7B2"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.7810 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 496.2070 mm.

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Input, Nozzle Desc: N31"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		210.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	20.0000	mm.
Height of Beveled Transition	L'	12.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm.
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	483.4740	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N31"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000	in.
Actual Thickness Used in Calculation	0.315	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$



= 10.5192 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.5 \cdot 15.7 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
= 0.3283 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $11.0 / 0.5773$
= 19.0526 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	63.4000	mm.
Parallel to Vessel Wall	Rn+tn+t	31.7000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N31".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3283	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.5192	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5192	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.1200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[6.12, max(13.5192, 4.5)]
= 6.1200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.3283, 6.12)
= 6.1200 mm.

Available Nozzle Neck Thickness = 8.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N31"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0059 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 498.4799 mm.

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Input, Nozzle Desc: N16"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		490.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	403.9720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	280.0000	mm.
Thickness of Pad	te	14.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm.
Reinforcing Pad Width		55.8625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869	in.
Actual Thickness Used in Calculation	0.378	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 10.5192 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 77.54 / (118 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 1.9003 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 310.1452 mm.
 Parallel to Vessel Wall, opening length d 155.0726 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.514	NA	NA
Area in Shell	A1	0.736	NA	NA
Area in Nozzle Wall	A2	2.211	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.295	NA	NA
Area in Element	A5	15.642	NA	NA
TOTAL AREA AVAILABLE	Atot	20.884	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 248.7846 14.0000 mm.
 Based on given Pad Diameter: 280.0000 10.0885 mm.
 Based on Shell or Nozzle Thickness: 285.6701 9.6012 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (155.0726 \cdot 10.5192 \cdot 1.0 + 2 \cdot 6.6012 \cdot 10.5192 \cdot 1.0 \cdot (1 - 0.86))$
 $= 16.514 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 155.073(1.0 \cdot 11.0 - 1.0 \cdot 10.519) - 2 \cdot 6.601 \\ &\quad (1.0 \cdot 11.0 - 1.0 \cdot 10.5192) \cdot (1 - 0.855) \\ &= 0.736 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (6.6 - 1.9) \cdot 0.8551 \\ &= 2.211 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 12.0^2 \cdot 1.0 \\ &= 2.295 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp,DL) - (\text{Nozzle OD})) \cdot (\min(tp,Tlwp,te)) \cdot fr4 \\ &= (280.0 - 168.275) \cdot 14.0 \cdot 1.0 \\ &= 15.642 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.9003 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 13.5192 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 13.5192 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.22, \max(13.5192, 4.5)] \\ &= 9.2200 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.9003, 9.22) \\ &= 9.2200 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 9.6012 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 49.4,	Allowable	: 118.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 245.6 N./mm2	Passed
Occasional	: 16.1,	Allowable	: 156.9 N./mm2	Passed
Shear	: 23.3,	Allowable	: 82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C



Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 1.9, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.288$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 1.9, c = 3.0 mm., $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.288$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-72 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $28.50 / 51.00 = 0.559$

Weld Size Calculations, Description: N16"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.6012 mm.
Intermediate Calc. for pad/shell Welds	TminPad	11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.6208 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (16.5136 - 0.7364 + 2 \cdot 6.6012 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 10.5192)) \cdot 138)$
 $= 22277.42 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (2.2108 + 15.6415 + 2.2951 - 0.0 \cdot 0.86) \cdot 138$
 $= 28350.55 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (2.2108 + 0.0 + 0.8551 + (1.2418)) \cdot 138$
 $= 6061.62 \text{ Kg}$



Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S$$

$$= (2.2108 + 0.0 + 2.2951 + 15.6415 + (1.2418)) \cdot 138$$

$$= 30097.97 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 15584. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 280.0 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 36391. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 80.8369) \cdot (9.6012 - 3.0) \cdot 0.7 \cdot 118$$

$$= 14120. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 168.275 \cdot 14.0 \cdot 0.74 \cdot 138$$

$$= 38534. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 30277. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (36391 + 14120) = 50511 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15584 + 38534 + 30277 + 0) = 84395 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (36391 + 30277 + 0) = 66668 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 50511	Kgf, must exceed W = 22277 Kgf	or W1 = 28350 Kgf
Path 2-2 = 84395	Kgf, must exceed W = 22277 Kgf	or W2 = 6061 Kgf
Path 3-3 = 66667	Kgf, must exceed W = 22277 Kgf	or W3 = 30097 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 7.1299 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 425.1020 mm.

Input Echo, WRC297 Item 1, Description: N16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1000.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2



Vessel Hot S.I. Allowable Smh 138.00 N./mm²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	149.073	mm.
Nozzle Thickness	9.6012	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	280.000	mm.
Thickness of Reinforcing Pad	14.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	28.50	bars
Radial Load	P	-407.91	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	275.24	Kg-m.
Longitudinal Moment	Ml	366.98	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(515.5(14.0 - 3.0))}$
 $= 416.774 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1031.000	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	25.000	mm.
Nozzle Thickness used	(t) =	6.601	mm.
T / t	=	3.787	
d / t	=	25.492	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.048$$

Nr/P	=	0.081
Mr/P	=	0.082
M0/P	=	0.105
N0/P	=	0.252
MrD/Mc	=	0.142
NrDL/Mc	=	0.119
M0d/Mc	=	0.223
N0DL/Mc	=	0.151
MrD/Ml	=	0.118
NrDL/Ml	=	0.104
M0D/Ml	=	0.174
N0DL/Ml	=	0.330

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	55	58	55	58

Outplane Stress Summary 14 80 108 32

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-24	24	24	-24
Inplane Pressure Stress	27	27	27	27

Inplane Stress Summary 3 45 59 4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4

Shear Stress Summary 4 4 3 3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	16	81	108	32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27

Outplane Stress Summary -4 55 71 -4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-21	21	21	-21



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	58	55	58
Inplane Stress Summary	34	74	84	37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	39	75	86	43

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	38	41	38	41
Hoop Stress Summary	28	32	51	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-193	193	193	-193
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-177	163	263	-167

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	15	15	15	15
Shear Stress Summary	18	18	11	11

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	209	166	264	223

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	38	41	38	41
Hoop Stress Summary	36	39	44	47

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-180	180	180	-180
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-158	158	244	-162

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	15	15	15	15
Shear Stress Summary	11	11	18	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	196	159	246	213

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(508.5(14.0 - 3.0))}$
 $= 415.081 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1017.000 mm.
 Nozzle Outside Diameter (d) = 280.000 mm.
 Vessel Thickness used (T) = 11.000 mm.
 Nozzle Thickness used (t) = 62.464 mm.
 $T / t = 0.176$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.647$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.063
 Mr/P = 0.118
 M0/P = 0.035



NØ/P = 0.062
 MrD/Mc = 0.276
 NrDL/Mc = 0.096
 MØd/Mc = 0.083
 NØDL/Mc = 0.079
 MrD/Ml = 0.091
 NrDL/Ml = 0.078
 MØD/Ml = 0.027
 NØDL/Ml = 0.080

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	128	131	128	131
Outplane Stress Summary	112	135	163	118

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	23	92	155	-6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	112	135	163	124

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	D1
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64

Outplane Stress Summary 27 92 119 26

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-131	131	131	-131
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	131	128	131
Inplane Stress Summary	14	234	293	-13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	27	234	294	40

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	55	58	55	58	55	58	55	58
Circ. Pl (SUS)	-10	-10	12	12	-3	-3	3	3
Circ. Q (SUS)	-31	31	39	-39	-18	18	24	-24
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-3	-3	3	3	-2	-2	4	4
Long. Q (SUS)	-21	21	27	-27	-30	30	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	55.0	58.0	55.0	58.0	55.0	58.0	55.0	58.0
Pm+Pl (SUS)	45.0	48.0	67.0	70.0	52.0	55.0	58.0	61.0
Pm+Pl+Q (Total)	15.3	79.5	106.3	31.6	39.8	73.8	83.1	44.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed

Pm+Pl+Q (TOTAL)	106.32	413.99	Passed
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Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	131	128	131	128	131	128	131
Circ. Pl (SUS)	-6	-6	10	10	-5	-5	9	9
Circ. Q (SUS)	-10	10	22	-22	-108	108	154	-154
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-6	-6	10	10	-4	-4	8	8
Long. Q (SUS)	-34	34	80	-80	-33	33	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	128.0	131.0	128.0	131.0	128.0	131.0	128.0	131.0
Pm+Pl (SUS)	122.0	125.0	138.0	141.0	123.0	126.0	137.0	140.0
Pm+Pl+Q (Total)	112.2	135.4	160.6	125.1	27.3	234.0	291.1	41.8

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	141.01	207.00	Passed
Pm+Pl+Q (TOTAL)	291.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	41	38	41	38	41	38	41
Circ. Pl (SUS)	-10	-10	12	12	-2	-2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-26	-26	28	28	-19	-19	21	21
Long. Q (SUS)	-171	171	215	-215	-158	158	202	-202
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	15	15	15	15	15	15	15	15
Pm (SUS)	38.0	41.0	38.0	41.0	38.0	41.0	38.0	41.0
Pm+Pl (SUS)	35.5	38.5	51.9	54.2	36.5	39.5	44.2	46.4
Pm+Pl+Q (Total)	209.1	166.4	262.7	222.3	195.5	159.2	243.6	210.1

Nozzle Stress Summation Comparison (N./mm2):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	118.00	Passed
Pm+Pl (SUS)	54.24	177.00	Passed
Pm+Pl+Q (TOTAL)	262.68	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N26"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2280.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	403.9720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	280.0000	mm.
Thickness of Pad	te	14.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm.
Reinforcing Pad Width		55.8625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N26"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869	in.
Actual Thickness Used in Calculation	0.378	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 10.5192 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 77.54 / (118 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 1.9003 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 310.1452 mm.
 Parallel to Vessel Wall, opening length d 155.0726 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.514	NA	NA
Area in Shell	A1	0.736	NA	NA
Area in Nozzle Wall	A2	2.211	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.295	NA	NA
Area in Element	A5	15.642	NA	NA
TOTAL AREA AVAILABLE	Atot	20.884	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 248.7846 14.0000 mm.
 Based on given Pad Diameter: 280.0000 10.0885 mm.
 Based on Shell or Nozzle Thickness: 285.6701 9.6012 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (155.0726 \cdot 10.5192 \cdot 1.0 + 2 \cdot 6.6012 \cdot 10.5192 \cdot 1.0 \cdot (1 - 0.86))$
 $= 16.514 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 155.073(1.0 \cdot 11.0 - 1.0 \cdot 10.519) - 2 \cdot 6.601 \\ &\quad (1.0 \cdot 11.0 - 1.0 \cdot 10.5192) \cdot (1 - 0.855) \\ &= 0.736 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (6.6 - 1.9) \cdot 0.8551 \\ &= 2.211 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 12.0^2 \cdot 1.0 \\ &= 2.295 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp,DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (280.0 - 168.275) \cdot 14.0 \cdot 1.0 \\ &= 15.642 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.9003 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 13.5192 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 13.5192 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.22, \max(13.5192, 4.5)] \\ &= 9.2200 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.9003, 9.22) \\ &= 9.2200 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 9.6012 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 49.4,	Allowable	: 118.0 N./mm2	Passed
Expansion	: 0.0,	Allowable	: 245.6 N./mm2	Passed
Occasional	: 16.1,	Allowable	: 156.9 N./mm2	Passed
Shear	: 23.3,	Allowable	: 82.6 N./mm2	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C



Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 1.9, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.288$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 1.9, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.288$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-72 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $28.50 / 51.00 = 0.559$

Weld Size Calculations, Description: N26"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.6012 mm.
Intermediate Calc. for pad/shell Welds	TminPad	11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.6208 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$5.5000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (16.5136 - 0.7364 + 2 \cdot 6.6012 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 10.5192)) \cdot 138)$
 $= 22277.42 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (2.2108 + 15.6415 + 2.2951 - 0.0 \cdot 0.86) \cdot 138$
 $= 28350.55 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (2.2108 + 0.0 + 0.8551 + (1.2418)) \cdot 138$
 $= 6061.62 \text{ Kg}$



Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r 1)) \cdot S$$

$$= (2.2108 + 0.0 + 2.2951 + 15.6415 + (1.2418)) \cdot 138$$

$$= 30097.97 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 15584. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 280.0 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 36391. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 80.8369) \cdot (9.6012 - 3.0) \cdot 0.7 \cdot 118$$

$$= 14120. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 168.275 \cdot 14.0 \cdot 0.74 \cdot 138$$

$$= 38534. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 168.275 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 30277. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (36391 + 14120) = 50511 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (15584 + 38534 + 30277 + 0) = 84395 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (36391 + 30277 + 0) = 66668 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 50511	Kgf, must exceed W = 22277 Kgf	or W1 = 28350 Kgf
Path 2-2 = 84395	Kgf, must exceed W = 22277 Kgf	or W2 = 6061 Kgf
Path 3-3 = 66667	Kgf, must exceed W = 22277 Kgf	or W3 = 30097 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 7.1299 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 425.1020 mm.

Input Echo, WRC297 Item 1, Description: N26" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1000.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm2



Vessel Hot S.I. Allowable Smh 138.00 N./mm²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	149.073	mm.
Nozzle Thickness	9.6012	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-333 6	
Nozzle Cold S.I. Allowable	SNmc	118.00 N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00 N./mm ²

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	280.000	mm.
Thickness of Reinforcing Pad	14.0000	mm.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	28.50	bars
Radial Load	P	-407.91	Kgf
Circumferential Shear	Vc	611.87	Kgf
Longitudinal Shear	Vl	611.87	Kgf
Circumferential Moment	Mc	275.24	Kg-m.
Longitudinal Moment	Ml	366.98	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(515.5(14.0 - 3.0))}$
 $= 416.774 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1031.000	mm.
Nozzle Outside Diameter	(d) =	168.275	mm.
Vessel Thickness used	(T) =	25.000	mm.
Nozzle Thickness used	(t) =	6.601	mm.
T / t	=	3.787	
d / t	=	25.492	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.048$$

Nr/P	=	0.081
Mr/P	=	0.082
M0/P	=	0.105
N0/P	=	0.252
MrD/Mc	=	0.142
NrDL/Mc	=	0.119
M0d/Mc	=	0.223
N0DL/Mc	=	0.151
MrD/Ml	=	0.118
NrDL/Ml	=	0.104
M0D/Ml	=	0.174
N0DL/Ml	=	0.330

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-11	-11	11	11
Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	55	58	55	58
Outplane Stress Summary	14	80	108	32

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-24	24	24	-24
Inplane Pressure Stress	27	27	27	27
Inplane Stress Summary	3	45	59	4

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	4	4	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	16	81	108	32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-34	34	34	-34
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	27	27	27	27
Outplane Stress Summary	-4	55	71	-4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-21	21	21	-21



Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	55	58	55	58
Inplane Stress Summary	34	74	84	37

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	39	75	86	43

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-11	-11	11	11
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	38	41	38	41
Hoop Stress Summary	28	32	51	54

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-27	-27	27	27
Axial Bending (ML)	-193	193	193	-193
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-177	163	263	-167

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	15	15	15	15
Shear Stress Summary	18	18	11	11

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	B1
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	209	166	264	223



Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-3	-3	3	3
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	38	41	38	41
Hoop Stress Summary	36	39	44	47

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	22	-22	22	-22
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-180	180	180	-180
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	19	19	19	19
Axial Stress Summary	-158	158	244	-162

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	15	15	15	15
Shear Stress Summary	11	11	18	18

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	196	159	246	213

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 168.275 + 2 \cdot 1.65 \cdot \sqrt{(508.5(14.0 - 3.0))}$
 $= 415.081 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1017.000 mm.
 Nozzle Outside Diameter (d) = 280.000 mm.
 Vessel Thickness used (T) = 11.000 mm.
 Nozzle Thickness used (t) = 62.464 mm.
 $T / t = 0.176$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.647$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.063
 Mr/P = 0.118
 M0/P = 0.035



NØ/P = 0.062
 MrD/Mc = 0.276
 NrDL/Mc = 0.096
 MØd/Mc = 0.083
 NØDL/Mc = 0.079
 MrD/Ml = 0.091
 NrDL/Ml = 0.078
 MØD/Ml = 0.027
 NØDL/Ml = 0.080

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-8	-8	8	8
Outplane Bending (ML)	-16	16	16	-16
Normal Pressure Stress	128	131	128	131
Outplane Stress Summary	112	135	163	118

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-8	-8	8	8
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	23	92	155	-6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	112	135	163	124

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64

Outplane Stress Summary 27 92 119 26

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	23	-23	23	-23
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-131	131	131	-131
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	131	128	131
Inplane Stress Summary	14	234	293	-13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	27	234	294	40

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	55	58	55	58	55	58	55	58
Circ. Pl (SUS)	-10	-10	12	12	-3	-3	3	3
Circ. Q (SUS)	-31	31	39	-39	-18	18	24	-24
Long. Pm (SUS)	27	27	27	27	27	27	27	27
Long. Pl (SUS)	-3	-3	3	3	-2	-2	4	4
Long. Q (SUS)	-21	21	27	-27	-30	30	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	55.0	58.0	55.0	58.0	55.0	58.0	55.0	58.0
Pm+Pl (SUS)	45.0	48.0	67.0	70.0	52.0	55.0	58.0	61.0
Pm+Pl+Q (Total)	15.3	79.5	106.3	31.6	39.8	73.8	83.1	44.7

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	58.00	138.00	Passed
Pm+Pl (SUS)	70.00	207.00	Passed

Pm+Pl+Q (TOTAL)	106.32	413.99	Passed
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Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	131	128	131	128	131	128	131
Circ. Pl (SUS)	-6	-6	10	10	-5	-5	9	9
Circ. Q (SUS)	-10	10	22	-22	-108	108	154	-154
Long. Pm (SUS)	64	64	64	64	64	64	64	64
Long. Pl (SUS)	-6	-6	10	10	-4	-4	8	8
Long. Q (SUS)	-34	34	80	-80	-33	33	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	128.0	131.0	128.0	131.0	128.0	131.0	128.0	131.0
Pm+Pl (SUS)	122.0	125.0	138.0	141.0	123.0	126.0	137.0	140.0
Pm+Pl+Q (Total)	112.2	135.4	160.6	125.1	27.3	234.0	291.1	41.8

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+Pl (SUS)	141.01	207.00	Passed
Pm+Pl+Q (TOTAL)	291.09	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	38	41	38	41	38	41	38	41
Circ. Pl (SUS)	-10	-10	12	12	-2	-2	4	4
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	19	19	19	19	19	19	19	19
Long. Pl (SUS)	-26	-26	28	28	-19	-19	21	21
Long. Q (SUS)	-171	171	215	-215	-158	158	202	-202
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	15	15	15	15	15	15	15	15
Pm (SUS)	38.0	41.0	38.0	41.0	38.0	41.0	38.0	41.0
Pm+Pl (SUS)	35.5	38.5	51.9	54.2	36.5	39.5	44.2	46.4
Pm+Pl+Q (Total)	209.1	166.4	262.7	222.3	195.5	159.2	243.6	210.1

Nozzle Stress Summation Comparison (N./mm2):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	118.00	Passed
Pm+Pl (SUS)	54.24	177.00	Passed
Pm+Pl+Q (TOTAL)	262.68	354.00	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: N42"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2580.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		600	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N42"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 10.5192$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$



= 0.5939 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

Taking a UG-36(c)(3)(a) exemption for nozzle: N42".

This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5939	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.5192	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5192	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.8, max(13.5192, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.5939, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	13.9,	Allowable	:	138.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	331.1 N./mm ²	Passed
Occasional	:	2.4,	Allowable	:	183.5 N./mm ²	Passed
Shear	:	10.0,	Allowable	:	96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.956, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N42"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.0000	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.7810 mm.



The Cut Length for this Nozzle is, Drop + Ho + H + T : 212.6310 mm.

Input Echo, WRC297 Item 1, Description: N42" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	1000.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00 N./mm2
Vessel Hot S.I. Allowable	Smh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	50.800	mm.
Nozzle Thickness	16.7640	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-350 LF2	
Nozzle Cold S.I. Allowable	SNmc	138.00 N./mm2
Nozzle Hot S.I. Allowable	SNmh	138.00 N./mm2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	28.50	bars
Radial Load	P	-135.00	Kgf
Circumferential Shear	Vc	205.00	Kgf
Longitudinal Shear	Vl	205.00	Kgf
Circumferential Moment	Mc	35.00	Kg-m.
Longitudinal Moment	Ml	45.00	Kg-m.
Torsional Moment	Mt	55.00	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

*Warning - The ratio, Dn/Tn (5.030) is < 10
 check the limitations of WRC bulletin 297.*

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 84.328 + 2 \cdot 1.65 \cdot \sqrt{(508.5(14.0 - 3.0))}$
 $= 331.134 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1017.000 mm.
Nozzle Outside Diameter	(d) =	84.328 mm.
Vessel Thickness used	(T) =	11.000 mm.
Nozzle Thickness used	(t) =	13.764 mm.
T / t	=	0.799
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.797$$

Nr/P	=	0.132
Mr/P	=	0.233
M0/P	=	0.075
N0/P	=	0.139



MrD/Mc = 0.296
 NrDL/Mc = 0.122
 MØd/Mc = 0.106
 NØDL/Mc = -0.030
 MrD/Ml = 0.222
 NrDL/Ml = 0.111
 MØD/Ml = 0.081
 NØDL/Ml = 0.091

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	128	131	128	131
Outplane Stress Summary	110	145	160	111

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-57	57	57	-57
Inplane Pressure Stress	64	64	64	64
Inplane Stress Summary	18	103	143	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	110	146	160	113

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	1	1	-1	-1
Outplane Bending (Mc)	-21	21	21	-21
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	64	64	64	64



Outplane Stress Summary 50 83 91 38

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-59	59	59	-59
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	128	131	128	131
Inplane Stress Summary	81	173	209	62

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	82	173	209	63

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-3	-3	3	3
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	9	5	9
Hoop Stress Summary	3	6	11	14

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-9	-9	9	9
Axial Bending (ML)	-27	27	27	-27
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-26	14	47	-21

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1



Shear due to (V1)	0	0	0	0
Shear due to Torsion	3	3	3	3
Shear Stress Summary	4	4	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	31	16	47	36

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	1	1	-1	-1
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	9	5	9
Hoop Stress Summary	8	11	6	9

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-30	30	30	-30
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-27	19	47	-26

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (V1)	-1	-1	1	1
Shear due to Torsion	3	3	3	3
Shear Stress Summary	2	2	4	4

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	35	19	48	37

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	128	131	128	131	128	131	128	131
Circ. Pl (SUS)	-2	-2	4	4	-3	-3	5	5
Circ. Q (SUS)	-17	17	25	-25	-44	44	74	-74
Long. Pm (SUS)	64	64	64	64	64	64	64	64



Long. P1 (SUS)	-3	-3	5	5	2	2	0	0
Long. Q (SUS)	-42	42	72	-72	-17	17	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	128.0	131.0	128.0	131.0	128.0	131.0	128.0	131.0
Pm+P1 (SUS)	126.0	129.0	132.0	135.0	125.0	128.0	133.0	136.0
Pm+P1+Q (Total)	109.3	146.6	157.5	113.2	81.3	172.1	207.2	63.0

Vessel Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	131.00	138.00	Passed
Pm+P1 (SUS)	136.01	207.00	Passed
Pm+P1+Q (TOTAL)	207.21	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm2):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	9	5	9	5	9	5	9
Circ. P1 (SUS)	-2	-2	4	4	2	2	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. P1 (SUS)	-9	-9	9	9	-7	-7	7	7
Long. Q (SUS)	-21	21	33	-33	-24	24	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	5.0	9.0	5.0	9.0	5.0	9.0	5.0	9.0
Pm+P1 (SUS)	10.2	14.1	11.4	13.4	12.2	16.1	9.2	10.0
Pm+P1+Q (Total)	32.0	15.8	44.1	35.2	36.2	19.5	45.4	36.9

Nozzle Stress Summation Comparison (N./mm2):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	138.00	Passed
Pm+P1 (SUS)	16.12	207.00	Passed
Pm+P1+Q (TOTAL)	45.40	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: N51"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2000.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		1.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	8.0000	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		N/A	
Hub Height of Integral Nozzle	h	20.0000	mm.
Height of Beveled Transition	L'	12.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		20.0000	mm.
Style of FVC Nozzle		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	485.2520	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N51"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.000	in.
Actual Thickness Used in Calculation	0.315	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$



= 10.5192 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.5 \cdot 15.7 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
= 0.3283 mm.

Intermediate Hub Nozzle Calculations:

The Diameter Limit is less than the Hub OD !

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
= (Hub Thickness - Neck Thickness) / tan(30)
= $11.0 / 0.5773$
= 19.0526 mm.

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	63.4000	mm.
Parallel to Vessel Wall	Rn+tn+t	31.7000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N51".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.3283	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.5192	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5192	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 6.1200	mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.12, max(13.5192, 4.5)]
= 6.1200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.3283, 6.12)
= 6.1200 mm.

Available Nozzle Neck Thickness = 8.0000 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N51"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 1.0059 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 500.2579 mm.

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Input, Nozzle Desc: N86"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1410.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		6.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		80	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	403.9720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	280.0000	mm.
Thickness of Pad	te	14.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm.
Reinforcing Pad Width		55.8625	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N86"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	5.869	in.
Actual Thickness Used in Calculation	0.378	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 10.5192 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 77.54 / (118 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 1.9003 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 310.1452 mm.
 Parallel to Vessel Wall, opening length d 155.0726 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	16.514	NA	NA
Area in Shell	A1	0.736	NA	NA
Area in Nozzle Wall	A2	2.211	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.295	NA	NA
Area in Element	A5	15.642	NA	NA
TOTAL AREA AVAILABLE	Atot	20.884	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 248.7846 14.0000 mm.
 Based on given Pad Diameter: 280.0000 10.0885 mm.
 Based on Shell or Nozzle Thickness: 285.6701 9.6012 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (155.0726 \cdot 10.5192 \cdot 1.0 + 2 \cdot 6.6012 \cdot 10.5192 \cdot 1.0 \cdot (1 - 0.86))$
 $= 16.514 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 155.073(1.0 \cdot 11.0 - 1.0 \cdot 10.519) - 2 \cdot 6.601 \\ &\quad (1.0 \cdot 11.0 - 1.0 \cdot 10.5192) \cdot (1 - 0.855) \\ &= 0.736 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (6.6 - 1.9) \cdot 0.8551 \\ &= 2.211 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 12.0^2 \cdot 1.0 \\ &= 2.295 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (280.0 - 168.275) \cdot 14.0 \cdot 1.0 \\ &= 15.642 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.9003 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 13.5192 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 13.5192 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 9.2200 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[9.22, \max(13.5192, 4.5)] \\ &= 9.2200 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.9003, 9.22) \\ &= 9.2200 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 9.6012 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 9.601, tr = 1.9, c = 3.0 mm., E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.288, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.601, tr = 1.9, c = 3.0 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.288$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification	-46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2)	-72 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
 Design Pressure/Ambient Rating = $28.50/51.00 = 0.559$

Weld Size Calculations, Description: N86"

Intermediate Calc. for nozzle/shell Welds	Tmin	6.6012 mm.
Intermediate Calc. for pad/shell Welds	TminPad	11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.6208 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$5.5000 = 0.5 \cdot T_{minPad}$	$8.4840 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (16.5136 - 0.7364 + 2 \cdot 6.6012 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 10.5192)) \cdot 138)$
 $= 22277.42 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (2.2108 + 15.6415 + 2.2951 - 0.0 \cdot 0.86) \cdot 138$
 $= 28350.55 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (2.2108 + 0.0 + 0.8551 + (1.2418)) \cdot 138$
 $= 6061.62 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (2.2108 + 0.0 + 2.2951 + 15.6415 + (1.2418)) \cdot 138$
 $= 30097.97 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 15584. \text{ Kgf}$



Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 280.0 \cdot 12.0 \cdot 0.49 \cdot 138 \\ &= 36391. \text{ Kg} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 80.8369) \cdot (9.6012 - 3.0) \cdot 0.7 \cdot 118 \\ &= 14120. \text{ Kg} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 168.275 \cdot 14.0 \cdot 0.74 \cdot 138 \\ &= 38534. \text{ Kg} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 168.275 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 30277. \text{ Kg} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (36391 + 14120) = 50511 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (15584 + 38534 + 30277 + 0) = 84395 \text{ Kg} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (36391 + 30277 + 0) = 66668 \text{ Kg} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 50511	Kg, must exceed W = 22277 Kg	or W1 = 28350 Kg
Path 2-2 = 84395	Kg, must exceed W = 22277 Kg	or W2 = 6061 Kg
Path 3-3 = 66667	Kg, must exceed W = 22277 Kg	or W3 = 30097 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 7.1299 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 425.1020 mm.

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Input, Nozzle Desc: N92"

From: 20

Pressure for Reinforcement Calculations	P	28.560	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2550.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	206.3750	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N92"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.56 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.56)$
 $= 10.5417$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.56 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 28.56)$



= 0.5952 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N52".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5952	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.5417	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5417	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.8, max(13.5417, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.5952, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.542, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.958$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N92"

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 1.7810 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 222.1560 mm.

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Input, Nozzle Desc: N102"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1110.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		2.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	16.7640	mm.
Flange Material		SA-350 LF2 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	196.8500	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N102"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	2.000	in.
Actual Thickness Used in Calculation	0.660	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 10.5192$ mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 28.4 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$



= 0.5939 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	113.6000	mm.
Parallel to Vessel Wall, opening length	d	56.8000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N102".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.5939	mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000	mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.5192	mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.5192	mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000	mm.
Wall Thickness per table UG-45	tb3	= 7.8000	mm.

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.8, max(13.5192, 4.5)]
= 7.8000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(3.5939, 7.8)
= 7.8000 mm.

Available Nozzle Neck Thickness = 16.7640 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-35 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-37 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Weld Size Calculations, Description: N102"

Intermediate Calc. for nozzle/shell Welds	Tmin	11.0000	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 1.7810 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 212.6310 mm.

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Input, Nozzle Desc: N114"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1770.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	416.6720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	180.0000	mm.
Thickness of Pad	te	14.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm.
Reinforcing Pad Width		32.8500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N114"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.734	in.
Actual Thickness Used in Calculation	0.383	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 10.5192 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 50.42 / (118 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 1.2356 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 201.6618 mm.
 Parallel to Vessel Wall, opening length d 100.8309 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.812	NA	NA
Area in Shell	A1	0.475	NA	NA
Area in Nozzle Wall	A2	2.586	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.281	NA	NA
Area in Element	A5	9.198	NA	NA
TOTAL AREA AVAILABLE	Atot	14.541	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 153.3639 14.0000 mm.
 Based on given Pad Diameter: 180.0000 8.3241 mm.
 Based on Shell or Nozzle Thickness: 170.4807 9.7345 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (100.8309 \cdot 10.5192 \cdot 1.0 + 2 \cdot 6.7345 \cdot 10.5192 \cdot 1.0 \cdot (1 - 0.86))$
 $= 10.812 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 100.831(1.0 \cdot 11.0 - 1.0 \cdot 10.519) - 2 \cdot 6.735 \\ &\quad (1.0 \cdot 11.0 - 1.0 \cdot 10.5192) \cdot (1 - 0.855) \\ &= 0.475 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (6.73 - 1.24) \cdot 0.8551 \\ &= 2.586 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + (Wi - can/0.707)^2 - Ar \text{ Lost} \cdot fr2 + Trapfr4 \\ &= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 142.6332^2 \cdot 1.0 \\ &= 2.281 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (180.0 - 114.3) \cdot 14.0 \cdot 1.0 \\ &= 9.198 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2356 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 13.5192 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 13.5192 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.2578 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.258, \max(13.5192, 4.5)] \\ &= 8.2578 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.2356, 8.2578) \\ &= 8.2578 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\begin{aligned} &\text{Govrn. thk, } tg = 9.735, tr = 1.236, c = 3.0 \text{ mm.}, E* = 1.0 \\ &\text{Thickness Ratio} = tr \cdot E* / (tg - c) = 0.183, \text{Temp. Reduction} = 78 \text{ °C} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHt credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -35 °C
Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 1.236, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.183$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -46 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2) -72 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = $28.50/51.00 = 0.559$

Weld Size Calculations, Description: N114"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm.
Intermediate Calc. for pad/shell Welds TminPad 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.
Pad Weld	$5.5000 = 0.5 \cdot t_{minPad}$	$8.4840 = 0.7 \cdot W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (10.8119 - 0.4754 + 2 \cdot 6.7345 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 10.5192)) \cdot 138)$
 $= 14622.99$ Kg

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (2.5861 + 9.198 + 2.2814 - 0.0 \cdot 0.86) \cdot 138$
 $= 19792.44$ Kg

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (2.5861 + 0.0 + 0.8551 + (1.2669)) \cdot 138$
 $= 6625.06$ Kg

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$
 $= (2.5861 + 0.0 + 2.2814 + 9.198 + (1.2669)) \cdot 138$
 $= 21575.16$ Kg

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 10586$ Kg



Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 180.0 \cdot 12.0 \cdot 0.49 \cdot 138 \\ &= 23394. \text{ Kg} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 53.7827) \cdot (9.7345 - 3.0) \cdot 0.7 \cdot 118 \\ &= 9584. \text{ Kg} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 114.3 \cdot 14.0 \cdot 0.74 \cdot 138 \\ &= 26174. \text{ Kg} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 114.3 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 20565. \text{ Kg} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (23394 + 9584) = 32979 \text{ Kg}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (10586 + 26174 + 20565 + 0) = 57325 \text{ Kg} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (23394 + 20565 + 0) = 43960 \text{ Kg} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 32978	Kg, must exceed W = 14622 Kg	or W1 = 19792 Kg
Path 2-2 = 57325	Kg, must exceed W = 14622 Kg	or W2 = 6625 Kg
Path 3-3 = 43959	Kg, must exceed W = 14622 Kg	or W3 = 21575 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.2768 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.9489 mm.

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Input, Nozzle Desc: N124"

From: 20

Pressure for Reinforcement Calculations	P	28.500	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Cylindrical Shell	D	1000.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	cas	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		860.00	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-333 6 +IT	
Material UNS Number		K03006	
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	118.00	N./mm2
Allowable Stress At Ambient	Sna	118.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		4.0000	in.
Size and Thickness Basis		Minimum	
Nominal Thickness		120	
Flange Material		SA-350 LF2 +IT	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	416.6720	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm2
Pad Allowable Stress At Ambient	Spa	138.00	N./mm2
Diameter of Pad along vessel surface	Dp	180.0000	mm.
Thickness of Pad	te	14.0000	mm.
Weld leg size between Pad and Shell	Wp	12.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm.
Reinforcing Pad Width		32.8500	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: N124"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	3.734	in.
Actual Thickness Used in Calculation	0.383	in.

Nozzle input data check completed without errors.



Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 503.0 / (138 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 10.5192 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 28.5 \cdot 50.42 / (118 \cdot 1.0 - 0.6 \cdot 28.5)$
 $= 1.2356 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 201.6618 mm.
 Parallel to Vessel Wall, opening length d 100.8309 mm.
 Normal to Vessel Wall (Thickness Limit), pad side Tlwp 27.5000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	10.812	NA	NA
Area in Shell	A1	0.475	NA	NA
Area in Nozzle Wall	A2	2.586	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	2.281	NA	NA
Area in Element	A5	9.198	NA	NA
TOTAL AREA AVAILABLE	Atot	14.541	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 153.3639 14.0000 mm.
 Based on given Pad Diameter: 180.0000 8.3241 mm.
 Based on Shell or Nozzle Thickness: 170.4807 9.7345 mm.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (100.8309 \cdot 10.5192 \cdot 1.0 + 2 \cdot 6.7345 \cdot 10.5192 \cdot 1.0 \cdot (1 - 0.86))$
 $= 10.812 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:



$$\begin{aligned} &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\ &= 100.831(1.0 \cdot 11.0 - 1.0 \cdot 10.519) - 2 \cdot 6.735 \\ &\quad (1.0 \cdot 11.0 - 1.0 \cdot 10.5192) \cdot (1 - 0.855) \\ &= 0.475 \text{ cm}^2 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (6.73 - 1.24) \cdot 0.8551 \\ &= 2.586 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + (Wi - can/0.707)^2 - Ar \text{ Lost} \cdot fr2 + Trapfr4 \\ &= (1.) \cdot 0.86 + (0.0) \cdot 0.86 + 142.6332^2 \cdot 1.0 \\ &= 2.281 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (180.0 - 114.3) \cdot 14.0 \cdot 1.0 \\ &= 9.198 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2356 mm.
Wall Thickness per UG16(b),	tr16b = 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 13.5192 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 13.5192 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.5000 mm.
Wall Thickness per table UG-45	tb3 = 8.2578 mm.

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.258, \max(13.5192, 4.5)] \\ &= 8.2578 \text{ mm.} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(4.2356, 8.2578) \\ &= 8.2578 \text{ mm.} \end{aligned}$$

Available Nozzle Neck Thickness = 9.7345 mm. -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note:

This Material was specified as being an Impact Tested (Low Temperature) Material.

Impact Test Temperature provided per Specification	-46 °C
Calculated Minimum Design Metal Temperature	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\begin{aligned} &\text{Govrn. thk, } tg = 9.735, tr = 1.236, c = 3.0 \text{ mm.}, E* = 1.0 \\ &\text{Thickness Ratio} = tr \cdot E*/(tg - c) = 0.183, \text{Temp. Reduction} = 78 \text{ °C} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B



Govrn. thk, tg = 14.0, tr = 10.519, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.956$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -35 °C
Min Metal Temp. at Required thickness (UCS 66.1) -37 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 9.735, tr = 1.236, c = 3.0 mm., E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.183$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit -46 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -37 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -37 °C

Flange MDMT including Temperature reduction per UCS-66.1:

MDMT of ASME B16.5/47 flange per Matl. Specification -46 °C
Flange MDMT with Temp reduction per UCS-66(i)(2) -72 °C

Where the Stress Reduction Ratio per UCS-66(i)(2) is :
Design Pressure/Ambient Rating = $28.50 / 51.00 = 0.559$

Weld Size Calculations, Description: N124"

Intermediate Calc. for nozzle/shell Welds Tmin 6.7345 mm.
Intermediate Calc. for pad/shell Welds TminPad 11.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.7142 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm.
Pad Weld	$5.5000 = 0.5 * t_{minPad}$	$8.4840 = 0.7 * W_p$ mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (10.8119 - 0.4754 + 2 \cdot 6.7345 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 10.5192)) \cdot 138)$
 $= 14622.99$ Kg

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (2.5861 + 9.198 + 2.2814 - 0.0 \cdot 0.86) \cdot 138$
 $= 19792.44$ Kg

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (2.5861 + 0.0 + 0.8551 + (1.2669)) \cdot 138$
 $= 6625.06$ Kg

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (2.5861 + 0.0 + 2.2814 + 9.198 + (1.2669)) \cdot 138$
 $= 21575.16$ Kg

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 10586$ Kg



Shear, Pad Element Weld [Spew]:

$$\begin{aligned} &= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW \\ &= (3.1416/2.0) \cdot 180.0 \cdot 12.0 \cdot 0.49 \cdot 138 \\ &= 23394. \text{ Kgf} \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned} &= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn \\ &= (3.1416 \cdot 53.7827) \cdot (9.7345 - 3.0) \cdot 0.7 \cdot 118 \\ &= 9584. \text{ Kgf} \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg \\ &= (3.1416/2) \cdot 114.3 \cdot 14.0 \cdot 0.74 \cdot 138 \\ &= 26174. \text{ Kgf} \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned} &= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng \\ &= (3.1416/2.0) \cdot 114.3 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138 \\ &= 20565. \text{ Kgf} \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (23394 + 9584) = 32979 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned} &= (Sonw + Tpgw + Tngw + Sinw) \\ &= (10586 + 26174 + 20565 + 0) = 57325 \text{ Kgf} \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned} &= (Spew + Tngw + Sinw) \\ &= (23394 + 20565 + 0) = 43960 \text{ Kgf} \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 32978	Kgf, must exceed W = 14622 Kgf	or W1 = 19792 Kgf
Path 2-2 = 57325	Kgf, must exceed W = 14622 Kgf	or W2 = 6625 Kgf
Path 3-3 = 43959	Kgf, must exceed W = 14622 Kgf	or W3 = 21575 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 30 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 3.2768 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.9489 mm.

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Input, Nozzle Desc: N61 1/2"

From: 30

Pressure for Reinforcement Calculations	P	28.527	bars
Temperature for Internal Pressure	Temp	65	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm2
Parent Allowable Stress At Ambient	Sva	138.00	N./mm2
Inside Diameter of Elliptical Head	D	1000.00	mm.
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	14.0000	mm.
Head Internal Corrosion Allowance	cas	3.0000	mm.
Head External Corrosion Allowance	co	0.0000	mm.
Distance from Head Centerline	L1	360.0000	mm.
User Entered Minimum Design Metal Temperature		-38.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-350 LF2 +IT	
Material UNS Number		K03011	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm2
Allowable Stress At Ambient	Sna	138.00	N./mm2
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		1.5000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm.
Flange Material		SA-105 +IT	
Flange Type		Long Weld Neck	
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	207.9620	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Note : Checking Nozzle in the Meridional direction.

Reinforcement CALCULATION, Description: N61 1/2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500	in.
Actual Thickness Used in Calculation	0.394	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 $= P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 $= 28.53 \cdot 1006.0 \cdot 0.992 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 28.53)$
 $= 10.3380 \text{ mm.}$



Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.53 \cdot 22.05 / (138 \cdot 1.0 - 0.6 \cdot 28.53)$
= 0.4616 mm.

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	99.2228	mm.
Parallel to Vessel Wall, opening length	d	49.6114	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000	mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N61 1/2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 10.0$, $t_r = 0.462$, $c = 3.0$ mm., $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.066$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-44 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $28.53 / 51.00 = 0.559$

Weld Size Calculations, Description: N61 1/2"

Intermediate Calc. for nozzle/shell Welds	Tmin	7.0000	mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$4.9000 = 0.7 \cdot t_{min}$	$7.0700 = 0.7 \cdot W_o$ mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

Note : Checking Nozzle in the Latitudinal direction.

Reinforcement CALCULATION, Description: N61 1/2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.500	in.
Actual Thickness Used in Calculation	0.394	in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
= $P \cdot D \cdot K / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
= $28.53 \cdot 1006.0 \cdot 0.992 / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 28.53)$
= 10.3380 mm.

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $28.53 \cdot 22.05 / (138 \cdot 1.0 - 0.6 \cdot 28.53)$
= 0.4616 mm.



UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	88.2000 mm.
Parallel to Vessel Wall, opening length	d	44.1000 mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	17.5000 mm.

*Taking a UG-36(c)(3)(a) exemption for nozzle: N61 1/2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 3.4616 mm.
Wall Thickness per UG16(b),	tr16b	= 4.5000 mm.
Wall Thickness, shell/head, internal pressure	trb1	= 13.3380 mm.
Wall Thickness	tb1 = max(trb1, tr16b)	= 13.3380 mm.
Wall Thickness	tb2 = max(trb2, tr16b)	= 4.5000 mm.
Wall Thickness per table UG-45	tb3	= 6.4200 mm.

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[6.42, max(13.338, 4.5)]
= 6.4200 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(3.4616, 6.42)
= 6.4200 mm.

Available Nozzle Neck Thickness = 10.0000 mm. -> OK

Weld Size Calculations, Description: N61 1/2"

Intermediate Calc. for nozzle/shell welds	Tmin	7.0000 mm.
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.9000 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm.

*Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles
exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.*

The Drop for this Nozzle is : 13.1002 mm.
The Cut Length for this Nozzle is, Drop + Ho + H + T : 236.2808 mm.

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